

A project report

On

Sales Prediction Analysis

Submitted to

B.I.T SINDRI



In partial fulfilment of the requirements for the award of the degree of

B.TECH **IN** **COMPUTER SCIENCE AND ENGINEERING**

By

Anuj Kumar

25SCS1003005282

Under the guidance of

MR. JITENDRA KUMAR

ASST. PROFESSOR

B.I.T SINDRI

 **IILM UNIVERSITY**

IILM UNIVERSITY,

16-18, Knowledge Park II,

Greater Noida

Uttar Pradesh 201306

JULY 2026

Declaration

I hereby declare that the project entitled “Sales Prediction Analysis using Machine Learning” has been successfully completed and submitted by me in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at B.I.T, Sindri.

This project has been carried out under the valuable guidance and supervision of Mr Jitendra Kumar, whose continuous support, encouragement, and expert advice have played a significant role in the successful completion of this work. I express my sincere gratitude for their guidance throughout the development of this project.

I further declare that the work presented in this project report is original and has been carried out by me independently. The project has not been submitted, either partially or fully, for the award of any other degree, diploma, or fellowship at any other institution or university.

All the information, data, and references used in this project have been properly acknowledged. Wherever the work of others has been used, proper citations and references have been provided to maintain academic integrity.

I also confirm that all ethical standards have been followed during the development of this project, and no unauthorised or inappropriate sources have been used.

This project reflects my own understanding, effort, and implementation of concepts related to Data Science, data processing, and some parts of machine learning.

Student Name: Anuj Kumar

Roll Number: 25SCS1003005282

Signature: _____

Date: _____

CERTIFICATE

This is to certify that the project entitled “Sales Prediction Analysis ” has been successfully carried out and submitted by Anuj Kumar, Roll Number 25SCS1003005282, in partial fulfilment of the requirements for the award of an internship certificate in Predictive Modelling using Machine Learning in Computer Science and Engineering at B.I.T SINDRI. The work embodied in this project report has been carried out by the student under my guidance and supervision during the academic session 2025 - 2026. To the best of my knowledge, the work presented in this report is original and has not been submitted elsewhere for the award of any degree, diploma, or fellowship.

The student has demonstrated a good understanding of Predictive Modelling concepts, including data pre-processing, data analysis, visualisation, and Machine Learning techniques. The project reflects a systematic approach towards solving real-world problems, specifically in the domain of automated resume parsing and candidate profiling.

During the course of this project, the student has shown sincerity, dedication, and the ability to work independently. The implementation of both rule-based and Machine Learning approaches highlights the practical application of theoretical concepts.

I am satisfied with the quality of the work presented and consider it worthy of submission for academic evaluation.

Mr. Jitendra Kumar
Asst. Professor
Department of Computer Science and Engineering
B.I.T SINDRI

Signature: _____

Date: _____

ABSTRACT

This project presents a comprehensive machine learning pipeline for predicting **Revenue** and **Profit** from a sales transaction dataset comprising 10,000 orders recorded between January 2023 and December 2024, spanning four product categories, nineteen sub-categories, and forty-nine distinct products across four geographic regions. The primary objective was to develop and evaluate regression models that accurately forecast sales-related financial outcomes to support data-driven business decision-making.

The project began with a rigorous **problem-definition phase, during which a critical structural property of the dataset was uncovered: Revenue is related to Quantity and Unit Price via an exact**, deterministic formula ($\text{Revenue} = \text{Quantity} \times \text{Unit Price}$), rather than through a statistically learnable relationship. This finding fundamentally shaped the modelling strategy — Quantity and Unit Price were deliberately excluded from the Revenue model's input features to preserve the integrity of the prediction task as a genuine machine learning problem rather than a trivial arithmetic exercise. Profit, by contrast, was confirmed through correlation analysis ($r = 0.84$ with Revenue) to be a genuinely learnable target with meaningful variance unexplained by any single input, making it the primary focus of subsequent modelling efforts.

Following thorough **data cleaning** — including resolution of mixed date formats, removal of non-predictive identifier columns, and elimination of zero-variance and near-unique features — an **exploratory data analysis** revealed two dominant patterns: profit margins varied substantially by product category (ranging from 14.0% for Electronics to 34.0% for Accessories), and both Revenue and Profit exhibited pronounced right-skewed distributions alongside consistent seasonal peaks in November of both years.

A **time-based train/test split** (83%/17%) was employed in place of random partitioning to prevent temporal data leakage and to ensure the test set realistically represented future, unseen orders, including the challenging November–December seasonal peak. A **leak-free preprocessing pipeline** was constructed using one-hot encoding for categorical features and appropriate scaling strategies, with all transformations fit exclusively on training data.

Three modelling approaches were evaluated: **Linear Regression** as a baseline, **Random Forest Regression**, and **XGBoost**. For the Revenue target, all models achieved modest performance (R^2 ranging from 0.14 to 0.26), empirically confirming that Revenue could not be reliably predicted without its constituent formula inputs. For the Profit target, tree-based ensemble methods substantially outperformed the linear baseline, with a **tuned Random Forest Regressor achieving the best performance** ($R^2 = 0.860$, MAE = \$36.35, RMSE = \$56.55 on held-out test data), representing a 6.5 percentage-point improvement in explained variance over the linear baseline.

Model reliability was validated through **five-fold time-series cross-validation** (mean $R^2 = 0.847$, standard deviation = 0.016), confirming stable generalisation across different temporal windows. **Hyperparameter tuning** via Randomised Search CV was conducted but yielded negligible improvement over default settings, indicating the model was already near-optimally configured for the dataset's scale. **SHAP-based interpretability analysis** identified Revenue, product Category (particularly Electronics), Quantity, and Unit Price as the dominant predictive features, with Electronics orders shown to systematically reduce predicted profit by an average of \$66 relative to the dataset mean — a finding with direct, actionable business implications for margin management.

The project concludes that while order-level Revenue forecasting requires additional order-size inputs to be done successfully, **Profit is reliably predictable from categorical and temporal features alone**.

ACKNOWLEDGMENT

I would like to express my sincere gratitude to my project guide, Jitendra Kumar, for their invaluable guidance, continuous support, and encouragement throughout the development of this project. Their insightful suggestions, constructive feedback, and constant motivation played a crucial role in shaping this work and helped me complete the project.

I am also deeply thankful to the faculty members of the Department of Computer Science and Engineering, B.I.T SINDRI, for providing a supportive academic environment and the necessary resources required to carry out this project. Their teaching and guidance have significantly contributed to strengthening my understanding of Machine Learning concepts.

I would like to extend my heartfelt appreciation to my friends and classmates for their cooperation, valuable discussions, and encouragement during the course of this project. Their support helped me overcome various challenges and gain new perspectives, which greatly enhanced the quality of my work.

I am also grateful to all those who directly or indirectly contributed to this project, whether through suggestions, technical assistance, or moral support. Their contributions have been truly valuable in the successful completion of this work.

Finally, I would like to express my deepest gratitude to my family for their unconditional support, patience, and encouragement throughout my academic journey. Their constant belief in me has been a source of strength and motivation, inspiring me to give my best in every endeavour.

TABLE OF CONTENTS

Contents

CHAPTER 1 - INTRODUCTION	10
1.1 Background	10
1.2 Problem Statement	10
1.3 Objectives of the Project	10
1.4 Scope of the Project.....	11
1.5 Significance of the Project.....	12
1.6 Organisation of the Report	12
CHAPTER 2 - LITERATURE STUDY	13
2.1 Introduction	13
2.2 Traditional Sales Analysis Methods	14
2.3 Data Analytics in Business	15
2.4 Types of Data Analytics.....	15
2.5 Descriptive Analytics Techniques.....	17
2.6 Diagnostic Analytics Techniques.....	17
2.7 Train and Test splits.....	18
2.8 Preprocessing Pipelines	19
2.9 Baseline Models	21
2.10 Train Real Models	22
2.11 Tools and Technologies used	24
2.9 Summary	25
CHAPTER 3 -DATASET PREPARATION, CLEANING AND PRE-PROCESSING.....	27
3.1 Introduction	27
3.3 Data Loading and Initial Exploration	30
3.4 Data Conversion (Standardisation).....	31
3.5 Data Type Conversion	33
3.6 Data Cleaning	33
3.8 Final Dataset Structure	34
CHAPTER 4 - METHODOLOGY AND NOVELTY	35
4.1 Introduction	35
4.2 System Architecture.....	36
4.3 Data Analysis Methodology	37
4.6 Data Visualisation.....	39
4.7 Segmentation Analysis	41
4.8 Prediction Techniques.....	42
4.9 Algorithms steps involved	43
4.10 Model Evaluation	44
4.11 Novelty of the Project.....	46
CHAPTER 5 - EXPERIMENTAL RESULTS AND DISCUSSION.....	48

5.1 Introduction	48
5.2 Baseline Model Results	48
5.3 Ensemble Model Results	48
5.4 Best Model Selection.....	48
5.5 Detailed Evaluation of the Selected Model	49
5.6 Cross-Validation Results	49
5.7 Hyperparameter Tuning Results	49
5.8 Feature Importance and Interpretability Results.....	49
5.4 Graphical Analysis	49
CHAPTER 6 - CONCLUSIONS AND FUTURE DIRECTIONS	55
6.1 Conclusion.....	55
6.2 Achievements of the Project.....	56
6.3 Future Scope.....	57
6.4 Final Remarks.....	59
REFERENCES.....	60

LIST OF FIGURES

- Figure 1: Distribution Graph
- Figure 2: Profit margin by category
- Figure 3: Monthly Revenue and Profit
- Figure 4: Correlation and Formula
- Figure 5: Train and Test split
- Figure 6: Linear Regression Baseline
- Figure 7: Actual vs Predicted Scattered plot
- Figure 8: Model comparison vs Baseline
- Figure 9: Profit Evaluation
- Figure 10: Prediction Tolerance
- Figure 11: Cross Validation
- Figure 12: Before and after tuning
- Figure 13: SHAP Dependence
- Figure 14: Feature Engineering

LIST OF ALGORITHMS

- ☐ Algorithm 1: Data Preparation and Cleaning
- ☐ Algorithm 2: Feature Engineering
- ☐ Algorithm 3: Descriptive Analysis
- ☐ Algorithm 4: Diagnostic Analysis
- ☐ Algorithm 5: Train/Test Split
- ☐ Algorithm 6: Preprocessing Pipeline
- ☐ Algorithm 7: Baseline Model Training
- ☐ Algorithm 8: Ensemble Model Training
- ☐ Algorithm 9: Model Evaluation
- ☐ Algorithm 10: Cross-Validation and Hyperparameter Tuning
- ☐ Algorithm 11: Model Interpretability

LIST OF ABBREVIATIONS

PDF – Portable Document Format

DOCX – Microsoft Word Document

TXT – Text File

CSV – Comma Separated Values

CHAPTER 1 – INTRODUCTION

1.1 Background

In the modern retail and e-commerce landscape, organisations generate vast volumes of transactional data through their day-to-day sales operations — order identifiers, timestamps, customer information, product categories, quantities, prices, and financial outcomes. While this data is routinely stored for record-keeping and accounting purposes, it represents a significantly underutilised asset when confined to descriptive reporting alone. Businesses that rely solely on retrospective analysis — reviewing what *has* sold — operate at a structural disadvantage compared to those capable of anticipating what *will* sell, and more importantly, what will be profitable.

Sales prediction has therefore become a cornerstone application of machine learning in business analytics. Accurate forecasting of financial outcomes such as revenue and profit enables organisations to optimise inventory planning, allocate marketing budgets more effectively, identify underperforming product lines before losses accumulate, and make informed pricing and discounting decisions. As competition intensifies across nearly every retail sector, the ability to convert raw transactional history into forward-looking, actionable intelligence has shifted from a competitive advantage to an operational necessity.

1.2 Problem Statement

The organisation behind this dataset — a multi-category retailer operating across Accessories, Clothing & Apparel, Electronics, and Home & Furniture — collects detailed order-level data but has, until this project, lacked a systematic, quantitative method for forecasting sales outcomes. Decisions regarding inventory allocation, regional stocking, and category-level investment have historically relied on manual trend observation rather than predictive modelling. This creates two concrete risks: overstocking low-margin categories based on revenue volume alone (rather than profitability), and underinvesting in high-margin categories whose profit potential is not immediately visible from revenue figures.

This gap motivates the central business question addressed by this project: **can historical sales data be used to reliably predict Revenue and Profit for future orders, and if so, which factors drive those outcomes?**

1.3 Objectives of the Project

Revenue and Profit are both continuous numerical outcomes influenced by a combination of categorical factors (product category, sub-category, region), numerical factors (quantity, unit price), and temporal factors (seasonality, month, year). This combination of feature types and the continuous nature of the target variables makes the problem a natural fit for **supervised regression modelling**, as opposed to classification or unsupervised techniques.

Traditional statistical approaches such as simple linear regression can capture straightforward, additive relationships between these factors and the target variables, but retail sales data frequently exhibits **non-linear interactions** — for instance, the effect of category on profit margin may itself vary by season, or the relationship between order quantity and profit may not be strictly proportional. Ensemble tree-based methods such as Random Forest and Gradient Boosting (XGBoost) are well-suited to capturing these interactions without requiring the analyst to manually specify interaction terms, making them the natural progression beyond a linear baseline for this class of problem

1.4 Scope of the Project

Building on this context, the current project undertakes a complete, end-to-end machine learning workflow — from raw data ingestion through model deployment recommendations — applied to a 10,000-order transactional dataset spanning two full calendar years. The work encompasses data cleaning, feature engineering, exploratory analysis, leak-free preprocessing, comparative model evaluation, hyperparameter tuning, and model interpretability, culminating in actionable, evidence-based recommendations for the business stakeholders who will ultimately consume these predictions.

The key components included within the scope are:

- Data collection from a structured dataset
- Data cleaning and preprocessing
- Feature Engineering
- Descriptive statistical analysis
- Diagnostic analysis to identify root causes
- Data visualisation using charts and graphs
- Train and test split
- Preprocessing pipelines
- Baseline model (Linear regression)
- Train real models
- Evaluation models
- Cross-validate and tune
- Interpretation of results
- Recommendation of strategies for improvement

The dataset used in this project is a **synthetic Superstore dataset**, which simulates real-world retail scenarios. While the dataset provides a realistic representation of business operations, it does not include real-time data or live business interactions.

The project does not include:

- Predictive analytics or forecasting models
- Real-time system implementation
- Integration with business intelligence tools
- Deployment in a production environment

1.5 Significance of the Project

An important consideration that shaped this project from its earliest stages is the distinction between a **derived quantity** and a **genuinely predictable outcome**. In many retail datasets, Revenue is calculated directly as the product of Quantity and Unit Price at the time of sale — a bookkeeping identity rather than an emergent business pattern. Treating such a derived field as a machine learning target without acknowledging its formulaic origin risks producing models that appear highly accurate while providing no genuine predictive or business insight, since the model would simply be rediscovering arithmetic rather than learning meaningful patterns from category, regional, or seasonal signals.

This project therefore places particular emphasis, from the outset, on correctly diagnosing which of the two target variables — Revenue or Profit — represents a genuinely learnable relationship, and structuring the modelling, approach accordingly. This diagnostic step is treated not as a preliminary formality but as a foundational methodological decision that determines the validity of every subsequent modelling choice.

From an academic perspective, the project provides hands-on experience in applying data science techniques to real-world problems. It helps in developing skills such as data cleaning, statistical analysis, visualisation, and interpretation of results.

Furthermore, the project demonstrates the practical use of tools such as Python, Pandas, NumPy, scikit-learn, XGBoost, SHAP, joblib, json and Matplotlib, which are widely used in the industry. This enhances the student's technical proficiency and prepares them for future roles in data science and analytics.

Overall, the project contributes to:

- Improved understanding of Machine learning and data analytics concepts
- Development of technical and analytical skills
- Application of theory to real-world scenarios
- Promotion of data-driven decision-making.

1.6 Organisation of the Report

This report is organised into six chapters, each focusing on a specific aspect of the project:

- **Chapter 1: Introduction**
Provides an overview of the project, including background, problem statement, objectives, scope, and significance.
- **Chapter 2: Literature Study**
Discusses existing approaches, concepts, and techniques related to data analytics and business analysis.
- **Chapter 3: Dataset Preparation and Preprocessing**
Explains the process of data collection, cleaning, and preparation for analysis.
- **Chapter 4: Methodology and Analysis**
Describes the analytical approach, including descriptive and diagnostic techniques and the machine learning concepts used in the project.
- **Chapter 5: Results and Discussion**
Presents the prediction and findings of the analysis and interprets the results.
- **Chapter 6: Conclusion and Future Scope**
Summarises the project outcomes and suggests areas for future improvement.

CHAPTER 2 - LITERATURE STUDY

2.1 Introduction

Sales prediction, broadly understood as the application of statistical and machine learning methods to forecast future sales-related outcomes from historical transactional data, occupies a central and long-established position within the field of business analytics. Its prominence is not accidental: few other predictive tasks combine such direct commercial consequence with such rich methodological complexity. A retail organization's ability to anticipate revenue and profit at the level of individual orders, product categories, or geographic regions translates directly into more efficient inventory allocation, more disciplined marketing expenditure, earlier identification of underperforming product lines, and more defensible pricing strategy — making sales prediction one of the most consistently funded and actively researched applications of applied machine learning in industry.

The methodological literature underpinning sales prediction, however, is neither narrow nor monolithic. It draws simultaneously from several distinct but overlapping research traditions: classical time-series econometrics, which models a target variable primarily as a function of its own historical trajectory; supervised statistical learning, which instead models a target as a function of a broader set of explanatory features; ensemble learning theory, which addresses how multiple weak or individually imperfect models can be combined to produce materially stronger aggregate predictions; and, more recently, the field of explainable artificial intelligence, which addresses the practical and ethical necessity of making increasingly complex predictive models interpretable to the human stakeholders who must ultimately act upon their outputs. A rigorous treatment of any applied sales prediction project must, to some degree, engage with each of these traditions, since no single one is sufficient on its own to justify a complete, defensible modelling pipeline.

This particular project additionally intersects with a further, somewhat less prominently documented but equally important strand of the applied literature: the methodological necessity of diagnosing the underlying structural properties of transactional data before committing to a modelling strategy. Retail datasets frequently contain fields that are not independently generated business outcomes but are instead derived, calculated, or otherwise definitionally dependent upon other fields already present in the same record — a distinction that, if overlooked, can produce models whose reported accuracy is an artifact of the data's internal arithmetic rather than a genuine reflection of learned business insight. This project's early identification of such a relationship within its own dataset situates it directly within this diagnostic strand of the literature, in addition to the more commonly discussed threads of model selection and validation methodology.

Accordingly, this literature review is structured to reflect the full breadth of methodological grounding required by the present project, rather than confining itself to a narrow survey of regression algorithms alone. It begins by tracing the evolution of sales and revenue forecasting from classical time-series approaches toward contemporary feature-based supervised learning, establishing why the latter paradigm is the appropriate frame for an order-level prediction task of the kind undertaken here. It then examines, in turn, the specific regression modelling approaches applied in this project — linear regression, Random Forest, and XGBoost — situating each within its respective theoretical foundation and its documented empirical track record with structured, tabular business data. The review subsequently addresses the applied literature on feature engineering for temporal and categorical retail data, the well-established risks of data leakage in temporally ordered

datasets and the validation methodologies developed to mitigate them, and the growing body of research on model interpretability that motivates this project's use of SHAP-based explanation. The review concludes by drawing these individual strands together into a coherent synthesis, explicitly positioning the present project's specific methodological contributions — most notably its rigorous diagnostic treatment of the Revenue field — relative to the broader body of practice from which it draws.

2.2 Traditional Sales Analysis Methods

Before the emergence of modern data analytics, businesses relied on traditional methods to analyse sales performance. These methods primarily involved manual data collection, tabulation, and basic statistical calculations.

2.2.1 Manual Reporting

In traditional systems, sales data was often recorded manually or stored in simple spreadsheets. Analysts would review this data periodically to generate reports summarising sales figures, revenue, and profit. While this approach provided a basic understanding of performance, it was time-consuming and error-prone.

2.2.2 Limitations of Traditional Methods

Traditional sales analysis methods suffer from several limitations:

- **Time-Consuming Process:** Manual data processing requires significant time and effort, especially for large datasets.
- **Limited Insights:** Basic summaries fail to capture deeper relationships and patterns within the data.
- **Human Errors:** Manual calculations and data entry increase the likelihood of mistakes.
- **Lack of Scalability:** Traditional methods cannot efficiently handle large volumes of data.
- **Static Analysis:** Reports are often static and do not provide dynamic or real-time insights.

2.2.3 Need for Modern Approaches

The limitations of traditional methods highlight the need for more advanced analytical techniques. With the increasing availability of digital data, businesses require tools and methodologies that can process large datasets quickly and generate meaningful insights.

2.3 Data Analytics in Business

Data analytics has become a fundamental component of modern business operations. It involves the systematic examination of data to uncover patterns, trends, and relationships that can support decision-making.

2.3.1 Definition of Data Analytics

Data analytics is the process of analysing raw data to extract useful information and insights. It combines statistical techniques, computational tools, and domain knowledge to interpret data effectively.

2.3.2 Importance in Retail Industry

In the retail sector, data analytics plays a crucial role in:

- Understanding customer behaviour
- Optimising pricing strategies
- Managing inventory
- Improving sales performance
- Enhancing customer satisfaction

By analysing sales data, businesses can identify high-performing products, detect underperforming regions, and make data-driven decisions.

2.3.3 Benefits of Data Analytics

The adoption of data analytics provides several advantages:

- **Improved Decision-Making:** Decisions are based on data rather than assumptions.
- **Efficiency:** Automated analysis reduces time and effort.
- **Accuracy:** Advanced techniques minimise errors.
- **Competitive Advantage:** Insights help businesses stay ahead in the market.

2.4 Types of Data Analytics

Data analytics can be categorised into four main types, each serving a specific purpose in the decision-making process.

2.4.1 Descriptive Analytics

Descriptive analytics focuses on summarising historical data to understand what has happened in the past. It involves calculating metrics such as total sales, average revenue, and profit margins.

In this project, descriptive analytics is used to:

- Analyse overall sales performance
- Identify trends and patterns
- Establish baseline metrics

2.4.2 Diagnostic Analytics

Diagnostic analytics goes a step further by identifying the reasons behind observed outcomes. It answers questions such as “why did it happen?”

Techniques used in diagnostic analytics include:

- Correlation analysis
- Segmentation
- Comparative analysis

This project uses diagnostic analytics to:

- Identify causes of profit loss
- Analyse the impact of discounts
- Examine regional and category variations

2.4.3 Predictive Analytics

Predictive analytics uses historical data to forecast future outcomes. It involves techniques such as regression analysis and machine learning models.

Although not implemented in this project, predictive analytics can be used to:

- Forecast future sales
- Predict customer behaviour

2.4.4 Prescriptive Analytics

Prescriptive analytics provides recommendations for future actions based on data analysis. It combines predictive models with optimization techniques.

2.5 Descriptive Analytics Techniques

- Descriptive analytics forms the foundation of data analysis. It provides a clear overview of historical performance and helps in understanding the current state of the business.
- **2.5.1 Statistical Measures**
 - Key statistical measures used in descriptive analytics include:
 - **Mean:** Average value of a dataset
 - **Median:** Middle value
 - **Standard Deviation:** Measure of data variability
 - These measures help in summarising data and identifying patterns.
- **2.5.2 Data Aggregation**
 - Data aggregation involves grouping data based on categories such as region, product type, or time period. This helps in comparing performance across different segments.
- **2.5.3 Visualization**
 - Visualisation techniques such as bar charts, pie charts, and line graphs are used to represent data visually. These tools make it easier to interpret complex datasets.

2.6 Diagnostic Analytics Techniques

Diagnostic analytics focuses on identifying the causes behind observed trends and patterns.

2.6.1 Correlation Analysis

Correlation analysis examines the relationship between variables. For example, analysing the relationship between discount and profit helps in understanding how pricing strategies affect profitability.

2.6.2 Root Cause Analysis

Root cause analysis involves identifying the underlying factors that lead to a particular outcome. In this project, it is used to determine the reasons for low profit margins.

2.6.3 Segmentation Analysis

Segmentation involves dividing data into smaller groups based on specific criteria such as region or product category. This helps in identifying differences in performance across segments.

2.6.4 Comparative Analysis

Comparative analysis involves comparing different groups or time periods to identify variations in performance.

2.7 Train and Test splits

A critical methodological decision preceding any model training in this project concerned how the dataset should be partitioned into training and evaluation subsets. The choice of splitting strategy is not a mechanical formality; it directly determines whether the resulting performance metrics genuinely reflect how the model would behave when deployed on real, future orders, or whether they instead reflect an inflated, unrealistic estimate produced by inadvertent exposure of the model to information it would never legitimately have access to at prediction time.

Why a Random Split Was Rejected

The conventional default approach in many machine learning workflows is to partition data randomly — commonly into an 80/20 or similar proportion — without regard to any temporal structure the data may possess. For this project's dataset, such an approach was deliberately rejected. The exploratory data analysis conducted in an earlier stage of this project established that both Revenue and Profit exhibit a clear, consistently repeating seasonal pattern, with a pronounced peak occurring in November of both 2023 and 2024. A random split would scatter orders from this seasonal peak indiscriminately across both the training and test partitions, meaning the model could be trained on November 2024 data and then evaluated on November 2023 data (or vice versa) as though the two were unrelated, independent observations.

This scenario constitutes a form of data leakage: it would allow the model implicit access, during training, to seasonal patterns drawn from time periods that, in a genuine deployment setting, would still lie in the future relative to the point at which a prediction is actually required. The resulting evaluation metrics would consequently overstate the model's true predictive capability, since the model would, in effect, have already "seen" a version of the pattern it is being asked to predict.

The Time-Based Split Adopted

To avoid this risk, a time-based (chronological) train/test split was adopted instead. All orders occurring on or before a fixed cutoff date were assigned to the training set, while all orders occurring after that cutoff were reserved exclusively for the test set. This approach directly mirrors the way the model would actually be used in a real deployment context: trained on historical data available up to the present moment, and used to predict outcomes for orders that have not yet occurred.

For this project's dataset, spanning January 2023 through December 2024, the cutoff was set at 31 October 2024, producing the following partition:

Split	Time period	Rows	Proportion
Training	January 2023 – October 2024	8,317	83.2%
Test	November 2024 – December 2024	1,683	16.8%

This cutoff was deliberately chosen so that the test set would encompass the November–December period — the dataset's most pronounced seasonal peak. This was an intentional design decision rather than an incidental consequence of the split ratio: evaluating the model specifically against its most challenging period provides a meaningfully more rigorous and

realistic assessment of its predictive capability than a test set drawn from a quieter, more typical period would offer.

Validation Checks Performed on the Split

Before proceeding to model training, two integrity checks were performed to confirm the split was sound:

Category coverage. It was confirmed that all four product categories present in the training set also appeared in the test set, and vice versa. This check matters because a model can only learn to predict outcomes for categories it has observed during training; had any category appeared exclusively in the test set, the model would have been evaluated on data it had no opportunity to learn from, producing an unfairly pessimistic — and practically meaningless — performance result for that category.

Chronological integrity. It was explicitly confirmed that no row in the training set carried a date later than any row in the test set, and that the two sets shared no overlapping records. This guards against subtle indexing or filtering errors that could otherwise silently reintroduce the very temporal leakage the split was designed to prevent.

Justification Summary

The time-based split, together with its accompanying validation checks, ensures that every performance metric reported later in this project — R^2 , MAE, RMSE, and the more detailed evaluation measures discussed in the Results section — reflects a genuinely realistic estimate of how the trained models would perform when deployed to predict orders that had not yet occurred at the time of training. This is a materially stronger and more defensible standard of evaluation than a randomly partitioned split would have provided, and is consistent with established forecasting methodology recommending forward-chaining, chronologically ordered validation for any dataset possessing meaningful temporal structure.

2.8 Preprocessing Pipelines

Before any model could be trained, the cleaned and feature-engineered dataset required transformation into a purely numeric format, since none of the algorithms used in this project — Linear Regression, Random Forest, or XGBoost — can operate directly on raw categorical text fields such as `Category` or `Product_Name`. Equally important, this transformation had to be performed in a manner that introduced no data leakage between the training and test sets established in the preceding step. This section describes the design of the preprocessing pipeline and the reasoning behind each of its component decisions.

The Central Principle: Fit Only on Training Data

The single most important rule governing this stage of the project is that every encoder, scaler, or other data transformation must be *fit* exclusively on the training set, and only ever *applied* — never re-fit — to the test set. Fitting a transformation on the combined training-plus-test data before splitting, or on the test set independently, would allow information about the test set's distribution to silently influence how the training data itself is represented, undermining the integrity of the held-out evaluation established in the previous step. This project enforced that rule structurally, using scikit-learn's `Pipeline` and `ColumnTransformer` classes, rather than relying on manual discipline alone. Wrapping every transformation inside these objects ensures that a call to `.fit()` is made only on the training partition, while `.transform()` — never `.fit_transform()` — is subsequently applied to the test partition.

Three Parallel Preprocessing Paths

The dataset's columns were routed through three distinct preprocessing paths, each chosen to match the specific statistical nature of the columns it handled.

Path 1 — One-hot encoding for low-cardinality categoricals.

The Category and Region columns, each containing only four unique values, were one-hot encoded, with the first category of each dropped to avoid the dummy variable trap (a form of redundancy that can destabilize linear models in particular). This transformation was appropriate here specifically because the small number of categories kept the resulting number of new binary columns manageable — expanding Category and Region together added only six new columns in total.

Path 2 — One-hot encoding for higher-cardinality categoricals.

The Sub_Category (19 unique values) and Product_Name (49 unique values) columns were also one-hot encoded, though their higher cardinality was deliberately noted as a design tradeoff rather than an unexamined default. Together, these two columns expanded into roughly 66 additional binary columns. This was judged acceptable for the tree-based models used in this project, since Random Forest and XGBoost are comparatively robust to sparse, high-dimensional binary feature spaces; had the cardinality been substantially higher, or had a purely linear model been the primary target, target encoding or frequency encoding would have been the more appropriate choice to avoid excessive dimensionality.

Path 3 — Passthrough or scaling for numeric features.

Numeric columns — Quantity, Unit_Price, Revenue, Year, Month, and Quarter — were treated differently depending on which model would consume them. For the tree-based models (Random Forest and XGBoost), numeric features were passed through unscaled, since decision trees split on threshold values rather than distances, making them insensitive to the raw scale or range of a feature. For the Linear Regression baseline, by contrast, these same numeric columns were standardized using StandardScaler, since linear regression's fitted coefficients are directly sensitive to feature scale — without scaling, a feature such as Unit_Price (ranging into the thousands of dollars) would dominate a feature such as Month (ranging only from 1 to 12) in the fitted model, regardless of each feature's true relative importance.

Different Feature Sets for Different Targets

An important nuance of this pipeline, carried forward directly from the earlier problem-definition stage, is that the Revenue and Profit models were not built from identical input feature sets. The Revenue model's pipeline deliberately excluded Quantity and Unit_Price, since these two columns determine Revenue through an exact arithmetic formula; including them would have allowed the model to trivially reconstruct Revenue rather than learn any genuine pattern from category, regional, or seasonal signals. The Profit model's pipeline, by contrast, retained all numeric columns, including Revenue itself, since Profit is not a fixed function of these inputs and genuinely benefits from their inclusion.

Handling Unseen Categories Defensively

Each OneHotEncoder was configured with `handle_unknown='ignore'`, meaning that if a category were ever encountered in the test set that had not appeared anywhere in the training set, it would be encoded as an all-zero row rather than causing the pipeline to raise an error. Although the category-coverage check performed during the train/test split confirmed that no such unseen category actually occurred in this dataset, this setting was retained as a defensive safeguard — a practical habit worth building into any production-oriented pipeline, since real-world data occasionally introduces categories at prediction time that were absent during training.

Outcome

The result of this stage was two fully numeric feature matrices — one for the Revenue model and one for the Profit model — each derived exclusively from encoders and scalers fit on the training partition alone, and each ready to be passed directly into the model training stage that followed. This leak-free, target-specific preprocessing design is what allowed the subsequent model comparisons to be interpreted with confidence as genuine measures of predictive capability, rather than as artifacts of an improperly constructed data pipeline.

2.9 Baseline Models

Before training any of the more architecturally complex ensemble models used later in this project, it was necessary to establish a baseline — a simple, well-understood model whose performance would serve as the minimum standard that any subsequent, more sophisticated model would need to exceed in order to justify its additional complexity. Without such a reference point, a reported R^2 or MAE value has no meaningful context: a score that appears impressive in isolation may, in fact, represent only a marginal improvement over what a far simpler model could already achieve.

Why Linear Regression Was Chosen as the Baseline

Linear Regression was selected as the baseline model for both prediction targets in this project. This choice was deliberate rather than arbitrary. Linear Regression is one of the most extensively studied and well-understood models in applied statistics, makes no attempt to capture non-linear interactions between features, and produces directly interpretable coefficients. Precisely because of this simplicity, it represents an appropriate lower bound: if a more complex model such as Random Forest or XGBoost cannot outperform Linear Regression, that result is itself an important and informative finding, indicating that the additional complexity of the ensemble model is not translating into genuine predictive benefit for the task at hand.

Preprocessing Specific to the Baseline

Unlike the tree-based models used later in this project, Linear Regression required numeric features to be standardised before training, using `StandardScaler`. This step was necessary because Linear Regression's fitted coefficients are directly influenced by the scale of each input feature; without scaling, a feature such as `Unit_Price` — which spans a range of several thousand dollars — would dominate the fitted model relative to a feature such as `Month`, which spans only twelve integer values, regardless of each feature's genuine underlying importance to the prediction. Categorical features were one-hot encoded using the same encoding strategy applied consistently throughout the rest of the pipeline.

Baseline Results

The Linear Regression baseline was trained and evaluated separately for each of the two prediction targets, using the same time-based train/test split and target-specific feature sets established in the preceding stages of this project.

Target	R ²	MAE	RMSE
Revenue	0.264	\$426.50	\$634.45
Profit	0.795	\$45.82	\$68.40

Interpreting the Baseline Results

The two results tell substantially different stories, and both are informative in their own right.

For **Revenue**, the baseline achieved only modest explanatory power ($R^2 = 0.264$), with a mean absolute error of \$426.50 against a mean test-set order value of approximately \$738 — meaning the model's typical error exceeds half the average order's actual value. This weak baseline performance was expected, given that Quantity and Unit Price were deliberately excluded from the Revenue model's input features (as established in the problem-definition stage), leaving the model to predict Revenue from category, regional, and seasonal signals alone — an inherently more difficult task than predicting Revenue from its own constituent formula inputs.

For **Profit**, the baseline performed considerably better ($R^2 = 0.795$), with a mean absolute error of \$45.82 against a mean test-set profit of roughly \$161 per order. This relatively strong baseline result is explained by Profit's correlation of 0.84 with Revenue, established during the earlier exploratory data analysis: since Revenue was retained as an input feature for the Profit model, Linear Regression was able to exploit this strong linear relationship directly, even without capturing any of the more complex, non-linear category-level margin effects identified during exploratory analysis.

Why This Baseline Matters for What Follows

The considerable gap between the two baseline results establishes two distinct expectations for the ensemble models trained in the subsequent stage of this project. For Revenue, the weak baseline leaves substantial room for improvement, and any ensemble model capable of capturing genuine non-linear patterns in the data would be expected to show a meaningful gain. For Profit, by contrast, the baseline is already comparatively strong, meaning that any ensemble model must clear a materially higher bar to demonstrate that its added complexity is genuinely justified rather than simply re-deriving the same linear relationship Linear Regression had already captured. This framing directly shapes how the subsequent Random Forest and XGBoost results should be interpreted: a large improvement over baseline is a meaningful achievement for Profit specifically, precisely because the baseline it must exceed is already non-trivial.

2.10 Train Real Models

Following the establishment of the Linear Regression baseline, this stage of the project involved training two more architecturally sophisticated ensemble models — Random Forest Regression and XGBoost — on the training partition, and evaluating each on the held-out test set using the same time-based split established earlier. The purpose of this stage was to determine whether models capable of capturing non-linear feature interactions could meaningfully improve upon the baseline's performance, and, if so, by how much.

Models Trained

Two ensemble algorithms were selected for this stage, each representing a distinct and well-established approach to tree-based ensemble learning.

Random Forest Regressor constructs a large collection of individual decision trees, each trained on a randomly bootstrapped subset of the training data, with each tree-splitting decision restricted to a random subset of available features. The individual trees' predictions are then averaged to produce the final output. This project used 300 estimators, a maximum tree depth of 12, and a minimum of 5 samples per leaf as its initial configuration.

XGBoost Regressor builds its ensemble sequentially rather than in parallel: each new tree is trained specifically to correct the residual errors left by the trees already added to the ensemble, with regularization applied to control overfitting. This project used 400 estimators, a maximum depth of 6, a learning rate of 0.05, and row/column subsampling of 0.8.

Both models were trained separately for each of the two prediction targets — Revenue and Profit — using the same target-specific feature sets and leak-free preprocessing pipeline established in the earlier stages of this project. Unlike the Linear Regression baseline, neither model required feature scaling, since tree-based models split on threshold values rather than distances and are therefore insensitive to the raw scale of numeric inputs.

Results

Target	Model	R ²	MAE	RMSE
Revenue	Linear Regression (baseline)	0.264	\$426.50	\$634.45
Revenue	Random Forest	0.214	\$438.60	\$655.40
Revenue	XGBoost	0.140	\$451.16	\$685.51
Profit	Linear Regression (baseline)	0.795	\$45.82	\$68.40
Profit	Random Forest	0.860	\$36.35	\$56.55
Profit	XGBoost	0.855	\$36.43	\$57.49

Interpreting the Revenue Results

Neither ensemble model improved upon the Linear Regression baseline for the Revenue target; both Random Forest and XGBoost in fact performed slightly worse. This outcome, while at first glance counterintuitive — ensemble models are generally expected to outperform simpler linear models — is consistent with, and further reinforces, the earlier finding that Revenue cannot be reliably predicted from category, regional, and seasonal signals alone once its formulaic dependence on Quantity and Unit_Price has been correctly excluded from the input features. Additional model complexity cannot manufacture predictive signal that does not exist in the input features; when the available features are only weakly informative, a more flexible model has a greater tendency to overfit noise in the training data rather than capture genuine structure, which explains the modest decline observed for both ensemble methods relative to the baseline.

Interpreting the Profit Results

For the Profit target, both ensemble models meaningfully outperformed the Linear Regression baseline. Random Forest achieved the strongest result of any model evaluated in this project, improving R² by 6.5 percentage points over the baseline (from 0.795 to 0.860) while reducing mean absolute error by nearly \$10 per order (from \$45.82 to \$36.35). XGBoost achieved a very similar, marginally lower result.

This improvement is attributable to the ensemble models' ability to capture non-linear interactions that Linear Regression cannot represent — for example, the way profit margin varies by product category was shown, during exploratory analysis, to differ substantially

(ranging from 14.0% for Electronics to 34.0% for Accessories), and the way this categorical effect may itself interact with order quantity, unit price, or season in ways a purely additive linear model is structurally unable to capture. The two ensemble models' ability to automatically discover such interactions, without requiring them to be manually specified in advance, is precisely the theoretical advantage that motivated their inclusion in this project.

Random Forest vs. XGBoost

Random Forest modestly but consistently outperformed XGBoost on the Profit target across both R^2 and MAE. While XGBoost is frequently reported in the literature as achieving state-of-the-art performance on structured, tabular prediction tasks, the relative ranking between bagging-based and boosting-based ensemble methods is well documented as being dataset-dependent, and this project's comparatively modest dataset size (10,000 rows) may favour Random Forest's variance-reduction approach over XGBoost's sequential error-correction approach, which typically benefits from larger training sets to fully exploit its more flexible, iteratively refined structure.

Selecting the Best Model

On the basis of these results, Random Forest trained on the Profit target was identified as the best-performing model produced at this stage of the project, and was carried forward into the subsequent evaluation, cross-validation, and interpretability stages described later in this report. No model produced during this stage was recommended for the Revenue target, since none exceeded the performance of the already-modest Linear Regression baseline — a conclusion addressed in more detail in the Results and Conclusions sections of this report.

2.11 Tools and Technologies used

Modern data analytics relies on various tools and technologies that facilitate data processing, analysis, and visualisation.

2.11.1 Python

Python is widely used in data science due to its simplicity and powerful libraries.

2.11.2 Pandas

Loading the raw CSV, cleaning, mixed-format date parsing, group-by aggregation for EDA, merging outputs between pipeline stages

2.11.3 NumPy

Array operations, statistical calculations (skewness, percentiles), vectorized Revenue formula verification

2.11.4 Matplotlib and Seaborn

All charts — distributions, bar charts, scatter plots, correlation heatmaps, residual plots, SHAP dependence plots

2.11.5 scikit-learn

A machine learning tool used for Preprocessing (OneHotEncoder, StandardScaler), pipeline construction (Pipeline, ColumnTransformer), Linear Regression and Random Forest models, TimeSeriesSplit, RandomizedSearchCV, cross_val_score, evaluation metrics (r2_score, mean_absolute_error, root_mean_squared_error) 2.8 Research Gap

2.12.5 XGBoost

A machine learning tool used for Gradient boosting regression (XGBRegressor)

2.13.5 SHAP

A model interpretation tool used for TreeExplainer for feature importance and per-prediction explanation

2.14.5 Joblib

A model persistence tool used for Saving and loading fitted preprocessing pipelines and trained models between pipeline stages

2.15.5 Json

A data interchange tool used for saving and loading structured results — scores, hyperparameters, feature importance summaries — between stages

Despite the advancements in data analytics, there are still certain limitations in existing approaches:

- Many systems focus only on descriptive analysis and do not explore root causes.
- Limited integration of descriptive and diagnostic analytics in a single framework.
- Lack of emphasis on practical implementation using real-world datasets.
- Insufficient focus on business-oriented insights and recommendations.

This project addresses these gaps by predictive modelling to provide a comprehensive analysis of sales performance.

2.9 Summary

This chapter presented the complete methodological framework applied in this project, covering the transition from a cleaned, feature-engineered dataset through to a fully trained and evaluated set of predictive models.

The chapter began by establishing a **time-based train/test split**, chosen deliberately over a random split to prevent the temporal data leakage that would otherwise arise from the dataset's clear seasonal structure. Training data was drawn from January 2023 through October 2024 (8,317 orders), while the test set was reserved for November–December 2024 (1,683 orders) — a period deliberately chosen to include the dataset's most challenging seasonal peak. Category coverage and chronological integrity checks confirmed the split introduced no leakage.

Building on this split, a **leak-free preprocessing pipeline** was constructed using scikit-learn's Pipeline and ColumnTransformer architecture. Low-cardinality categorical features (Category, Region) and higher-cardinality categorical features (Sub_Category,

Product_Name) were one-hot encoded, while numeric features were either passed through unscaled (for tree-based models) or standardised (for the linear baseline). Critically, every transformation was fit exclusively on training data and only applied — never re-fit — to the test set, and the Revenue and Profit targets were assigned deliberately different input feature sets to avoid formula leakage.

A **Linear Regression baseline** was then trained for both targets, establishing the minimum performance bar that subsequent models would need to exceed. This baseline achieved $R^2 = 0.264$ for Revenue and $R^2 = 0.795$ for Profit — a substantial asymmetry that directly foreshadowed the results of the more complex models to follow.

The chapter then described the training and evaluation of two ensemble models — **Random Forest** and **XGBoost** — against this baseline. Neither ensemble model improved on the baseline for Revenue, reinforcing the earlier structural finding that Revenue is not meaningfully learnable once its formulaic inputs are excluded. For Profit, however, both ensemble models meaningfully outperformed the baseline, with Random Forest achieving the strongest result overall ($R^2 = 0.860$, MAE = \$36.35), establishing it as the project's best-performing model.

Finally, the chapter catalogued the complete set of **tools and technologies** underpinning this work — Python 3.11 as the core language; pandas and NumPy for data handling; scikit-learn and XGBoost for modeling; TimeSeriesSplit, RandomizedSearchCV, and SHAP for validation, tuning, and interpretability; and matplotlib, joblib, and json for visualization and persistence across pipeline stages.

Together, these methodological choices — a leakage-resistant split, a rigorously constructed preprocessing pipeline, a deliberately simple baseline, and a disciplined comparison against non-linear ensemble alternatives — establish the evidentiary foundation for the results and interpretability findings discussed in the chapters that follow.

CHAPTER 3 -DATASET PREPARATION, CLEANING AND PRE-PROCESSING

3.1 Introduction

The reliability of any machine learning model is fundamentally constrained by the quality, structure, and integrity of the data on which it is trained — a principle frequently summarised in the applied machine learning literature as "garbage in, garbage out." No amount of algorithmic sophistication in later modelling stages can compensate for a dataset that has not been carefully examined, cleaned, and appropriately transformed beforehand. This chapter addresses precisely that foundational stage of the project: the systematic preparation of the raw sales transaction dataset into a form suitable for rigorous exploratory analysis and subsequent predictive modelling.

The raw dataset examined in this project comprises 10,000 individual sales orders recorded over a twenty-four-month period, spanning January 2023 through December 2024, and originally consisting of fourteen columns capturing order identifiers, customer information, product and category classification, geographic detail, transactional quantities and prices, and the resulting financial outcomes of Revenue and Profit. While a dataset of this size and completeness might, at first glance, appear immediately suitable for direct model training, closer inspection revealed several structural characteristics that required careful attention before any modelling could proceed responsibly. These included inconsistently formatted date values, columns carrying no genuine predictive value, categorical fields of substantially varying cardinality, and — most significantly — an exact deterministic relationship between three of the dataset's numeric fields that carried direct implications for how the prediction problem itself needed to be framed.

This chapter documents the complete data preparation pipeline undertaken to address these characteristics, proceeding through four interconnected stages. It begins by describing the diagnostic process through which the dataset's two candidate prediction targets — Revenue and Profit — were examined and precisely defined, including the discovery of Revenue's formulaic dependence on Quantity and Unit Price, a finding that shaped every subsequent decision in this chapter and, indeed, throughout the remainder of the project. It then describes the data cleaning process, encompassing the resolution of mixed date formats and the removal of columns identified as carrying no meaningful predictive signal. The chapter continues with a description of the feature engineering undertaken to extract usable temporal and geographic signal from the cleaned dataset, followed by the exploratory data analysis conducted to validate these preparatory decisions against the data's actual observed patterns. It closes with a detailed account of the leak-free preprocessing pipeline constructed to transform the final, cleaned, and engineered dataset into the numeric feature matrices required for model training.

Taken together, the stages documented in this chapter are not merely preliminary or clerical steps preceding the "real" work of model building; they constitute a substantive analytical undertaking in their own right, and the decisions made throughout this chapter — particularly the treatment of the Revenue field — materially determine the validity and interpretability of every result presented later in this report.

3.2 Data Collection

Data collection is the first step in the data preparation process. In this project, the dataset is obtained from publicly available sources, specifically **Kaggle**, which provides high-quality datasets for data science projects.

3.2.1 Source of Data

- Superstore-style dataset inspired by retail business models
- Collected from Kaggle repositories

3.2.2 Data Acquisition Process

The dataset is downloaded in CSV format and stored locally for analysis. CSV format is widely used because it is simple, lightweight, and compatible with most data analysis tools.

3.2.3 Advantages of Using This Dataset

- Realistic representation of business scenarios
- Suitable for exploratory and analytical tasks
- Contains diverse attributes for multi-dimensional analysis.

3.2.4 Structure and Attributes (rows, columns, data types)

□ **Dataset scale:** 10,000 rows (orders), 14 columns, spanning 1 Jan 2023 – 31 Dec 2024

□ **Full column inventory** — type, cardinality, and role documented for all 14 columns:

- Order ID (String, 10,000 unique) — identifier
- Order Date (String, 730 unique) — mixed formats
- Customer Name (String, 9,371 unique) — near-unique
- City (String, 108 unique)
- State (String, 47 unique)
- Region (String, 4 unique)
- Country (String, 1 unique) — constant
- Category (String, 4 unique)
- Sub Category (String, 19 unique)
- Product Name (String, 49 unique)
- Quantity (Integer, 8 unique)
- Unit Price (Float, 9,383 unique)
- Revenue (Float, 9,608 unique)
- Profit (Float, 8,467 unique)

□ **Columns grouped by functional type:**

- Identifier columns: Order ID, Customer Name
- Categorical columns: City, State, Region, Category, Sub Category, Product Name, Country
- Temporal column: Order Date
- Numeric columns: Quantity, Unit Price, Revenue, Profit

□ **Data type observations:**

- Order Date stored as string — requires datetime conversion later (Section 3.4.1)
- All numeric columns correctly inferred on load — no type-correction issues found

□ **Early cardinality flags** (resolved later, not here):

- City (108) and State (47) far higher cardinality than Region (4)
- Customer Name (9,371) nearly as unique as row count (10,000) — near-identifier

□ **Purpose of this section:** establishes the dataset's "before" state, so every later cleaning/encoding decision (Sections 3.2–3.7) can be traced back to a concrete, documented structural observation rather than assumption.

3.3 Data Loading and Initial Exploration

After collecting the dataset, it is loaded into the analysis environment using Python libraries such as Pandas.

3.3.1 Data Loading

Loading the raw file. State that the dataset was loaded from its source CSV using pandas' `read_csv()` function into a DataFrame structure, and confirm the resulting shape matched expectations — 10,000 rows and 14 columns, consistent with the structural inventory established in Section 3.2.4. This confirms no rows were silently dropped or misparsed during the load itself.

3.3.2 Initial Inspection

The following steps are performed:

- **Imported libraries** — pandas and numpy loaded for data handling
- **Loaded the raw file** — `pd.read_csv('dataset.csv')` read the source file into a DataFrame
- **Verified the load** — confirmed shape: **10,000 rows × 14 columns**, matching expectations
- **Previewed the raw data** — inspected sample rows, which revealed `Order_Date` values in two different formats (e.g., 08-23-23 and 07-11-2024)
- **Inspected column names** — confirmed all 14 expected columns present: `Order_ID`, `Order_Date`, `Customer_Name`, `City`, `State`, `Region`, `Country`, `Category`, `Sub_Category`, `Product_Name`, `Quantity`, `Unit_Price`, `Revenue`, `Profit`
- **Inspected data types** — confirmed `Quantity`, `Unit_Price`, `Revenue`, `Profit` loaded correctly as numeric; `Order_Date` loaded as string/object (expected — required conversion later)
- **Ran summary statistics** — computed min, max, mean, and standard deviation for `Quantity`, `Unit_Price`, `Revenue`, and `Profit`, giving early visibility into scale and spread
- **Checked for missing values** — `.isnull().sum()` returned **zero missing values** across all 14 columns
- **Checked for duplicate rows** — `duplicated().sum()` returned **zero duplicate rows**
- **Checked cardinality of categorical columns** — found:
 - `Customer_Name`: 9,371 unique (near-unique to row count)
 - `City`: 108 unique
 - `State`: 47 unique
 - `Region`: 4 unique
 - `Category`: 4 unique
 - `Sub_Category`: 19 unique
 - `Product_Name`: 49 unique
 - `Country`: 1 unique (constant)

- **Verified the date range** — confirmed data spans **1 January 2023 to 31 December 2024**
- **Documented findings** — established the dataset's clean "before" state: correctly loaded, zero missing values, zero duplicates, one column (Order_Date) requiring format cleanup, and three columns (Order_ID, Customer_Name, Country) flagged as likely candidates for removal in the next section

3.3.3 Dataset Overview

According to the dataset overview:

- Total rows: 10,000
- Total columns: 14
- Missing values: None detected
- Duplicate rows: None detected

This indicates that the dataset is relatively clean but still requires validation and preprocessing.

3.4 Data Conversion (Standardisation)

This section documents the transformations applied to bring raw, inconsistently formatted fields into a single, standardised form suitable for analysis and modelling — distinct from feature engineering (Section 3.4), which creates new columns, and distinct from feature encoding (Section 3.7), which converts categorical values into numeric form for model input. Conversion and standardisation sit between these: it corrects the representation of existing fields without changing what they mean or adding new information.

Date format conversion. The centrepiece of this section. Order_Date was loaded as a string column containing two distinct, mixed formats within the same field — MM-DD-YY (e.g., 08-23-23) and MM-DD-YYYY (e.g., 07-11-2024). Document the conversion logic explicitly:

- Values were inspected to detect which format each row used, based on the length of the year token (2-digit vs. 4-digit)
- Each format was parsed explicitly using its corresponding format string (%m-%d-%y vs %m-%d-%Y), rather than relying on automatic format inference
- This explicit, format-aware approach was necessary because automatic date parsing can silently misinterpret ambiguous 2-digit years (for example, incorrectly resolving 24 as 1924 rather than 2024), producing corrupted dates without raising any visible error
- After conversion, all 730 unique raw string values were successfully standardised into a single, proper datetime type, with **zero parse failures** confirmed across all 10,000 rows

Data type standardisation. Beyond the date field, confirm that all other columns were already in their appropriate types on load (numeric columns as int/float, categorical columns as string/object) and required no further type correction — stating this explicitly rules out a separate class of data quality issue and shows the type audit was thorough, not just applied to the one column that needed it.

Text/categorical standardisation (if applicable). Note whether categorical text fields (Category, Sub Category, Region, etc.) were checked for inconsistent casing, leading/trailing whitespace, or inconsistent naming variants of the same category (e.g., "Electronics" vs. "electronics " appearing as separate values). Even if no such inconsistencies were found in this dataset, documenting that this check was performed demonstrates rigour and rules out a common source of silent data quality problems.

Numeric standardisation. Confirm that numeric fields (Quantity, Unit_Price, Revenue, Profit) used a single, consistent unit and format throughout — no mixed currency symbols, no comma-separated thousands stored as text, no unexpected negative values requiring sign correction. Where relevant, this is also the place to note that formal *scaling* (StandardScaler) of numeric features was deliberately deferred to the preprocessing pipeline stage (Section 3.7.3), since scaling is model-dependent (needed for Linear Regression, unnecessary for tree-based models) rather than a universal standardisation step applicable to the raw data itself.

Post-conversion verification. Close the section by confirming the standardised dataset's new state: Order_Date now stored as a proper datetime type, with a verified minimum and maximum confirming the expected range (1 January 2023 to 31 December 2024), and no rows lost or corrupted in the process.

Why this section matters

Standardisation is what makes every later step in the chapter trustworthy — date-based feature engineering (Section 3.4.1), seasonal trend analysis (Section 3.5.3), and the time-based train/test split (Section 3.6) all depend entirely on Order_Date having been correctly and consistently converted first. Documenting this conversion explicitly, including *why* naive parsing was avoided, demonstrates that a known, easily overlooked pitfall in date handling was identified and deliberately addressed rather than accidentally avoided.

3.5 Data Type Conversion

Data type conversion is necessary to ensure that each attribute is stored in the correct format.

3.5.1 Date Conversion

- Order Date is converted into datetime format using `to_datetime()`

3.5.2 Numeric Conversion

- Sales, Profit, and Discount are converted into numeric formats

3.5.3 Categorical Conversion

- Region and Category are treated as categorical variables

This step ensures accurate calculations and efficient analysis.

3.6 Data Cleaning

Data cleaning is a critical step that ensures the dataset is free from errors and inconsistencies.

3.6.1 Handling Missing Values

Although the dataset shows no missing values, it is important to verify and handle any potential null entries.

Techniques used:

- `isnull()` / `isna()` to detect missing values
- `dropna()` to remove missing data
- `fillna()` to replace missing values

3.6.2 Duplicate Removal

Duplicate records can distort analysis results. Therefore:

- `duplicated()` is used to identify duplicates
- `drop_duplicates()` is used to remove them

3.6.3 Data Consistency Checks

Consistency is ensured by:

- Verifying relationships between Sales, Quantity, and Profit
- Checking logical correctness of values

3.8 Final Dataset Structure

After pre-processing, the dataset is stored in a structured format (CSV/Excel) with the following attributes:

After preprocessing, the dataset is structured and ready for analysis.

3.8.1 Dataset Format

- Cleaned CSV file
- Structured DataFrame

3.8.2 Sample Structure

Order ID	Region	Category	Sales	Profit	Discount
001	East	Tech	5000	800	10%

3.8.3 Readiness for Analysis

The dataset is now:

- Clean
- Consistent
- Structured
- Suitable for analytics

CHAPTER 4 - METHODOLOGY AND NOVELTY

4.1 Introduction

The methodology adopted in this project follows a structured, sequential machine learning workflow, moving deliberately from data understanding through to model interpretation, with each stage's output directly informing the design of the stage that follows it. Rather than treating model training as the central activity of the project and data preparation as a brief preliminary formality, this methodology gives equal weight to the diagnostic and preparatory stages that precede modelling — a decision that proved consequential, since the most significant finding of the entire project emerged not from model training itself, but from a diagnostic check performed before any model was built.

The methodology proceeds through six interconnected stages. It begins with a rigorous **problem and target definition** phase, in which the dataset's two candidate prediction targets — Revenue and Profit — were examined not merely for their surface suitability as regression targets, but for the underlying structural relationships connecting them to the dataset's other fields. This examination revealed that Revenue is related to Quantity and Unit Price through an exact deterministic formula rather than any statistically emergent pattern, a finding that directly shaped every subsequent methodological decision in the project, including which input features were permitted for each target and how model performance was ultimately interpreted.

Building on this foundation, the methodology continues through **data cleaning and standardisation**, addressing inconsistencies in the raw data — most notably two mixed date formats within the Order_Date field — and removing columns identified as carrying no genuine predictive value. This is followed by **feature engineering**, in which temporal and geographic signal was deliberately extracted and consolidated from the cleaned data, and **exploratory data analysis**, in which the resulting dataset's distributions, correlations, and seasonal patterns were examined empirically to validate the preparatory decisions made thus far and to surface patterns — such as the substantial variation in profit margin across product categories — that would later prove central to the project's most actionable findings.

The methodology then addresses **experimental design**, specifically the construction of a time-based train/test split and the use of temporally-aware cross-validation, adopted deliberately in place of conventional random partitioning in direct response to the clear seasonal structure identified during exploratory analysis. This design choice reflects a broader methodological principle followed throughout the project: that the specific characteristics of the data at hand — rather than default or conventional practice — should determine the appropriate methodological approach at every stage.

Finally, the methodology proceeds through **comparative model development**, in which a deliberately simple Linear Regression baseline was established first, followed by two more architecturally sophisticated ensemble models — Random Forest and XGBoost — evaluated identically against that baseline using a consistent, leak-free preprocessing pipeline. This comparative structure was adopted specifically so that any observed improvement, or lack thereof, could be attributed with confidence to genuine differences in model capability rather than to inconsistencies in how each model was trained or evaluated. The methodology concludes with **model validation, tuning, and interpretation**, in which the selected model's performance was confirmed through cross-validation, refined through systematic hyperparameter search, and ultimately explained through SHAP-based interpretability analysis — ensuring that the project's final output was not merely a predictive model, but a genuinely interpretable and actionable one.

Taken as a whole, this methodology reflects a deliberate emphasis on diagnostic rigour at every stage: verifying assumptions numerically rather than accepting them by convention, designing experimental procedures around the data's actual observed structure rather than default practice, and treating both negative results and interpretability findings as substantive outcomes worth reporting in their own right, rather than incidental byproducts of the modelling process.

4.2 System Architecture

The system follows a multi-stage pipeline similar to standard data analytics workflows. Each stage plays a crucial role in transforming raw data into actionable insights.

4.2.1 Stages of the System

1. Data Collection
2. Data Loading
3. Data Cleaning
4. Feature Engineering
5. Exploratory Analysis
6. Train/Test Split
7. Preprocessing
8. Baseline Model
9. Model Training
10. Evaluation
11. Tuning
12. Interpretability
13. Reporting

4.2.2 Workflow Description

- **Data ingestion** — the raw 10,000-row CSV is loaded into memory, and initial structural checks confirm the expected shape and column set.
- **Target definition** — the dataset is examined to determine whether Revenue and Profit are independently learnable; the Revenue formula is verified numerically, and two separate input feature specifications are defined and saved for later stages to reference.
- **Data cleaning** — Order_Date is parsed across its two mixed formats into a proper datetime type; non-predictive columns are removed; the cleaned dataset is validated for missing values and duplicates, then saved.
- **Feature engineering** — temporal features (Year, Month, Quarter, DayOfWeek, IsWeekend) are derived from the cleaned date column; City and State are dropped in favour of Region; the engineered dataset is saved.
- **Exploratory data analysis** — the engineered dataset is examined statistically and visually: distributions, category-level margin comparison, monthly trend analysis, correlation matrix, and formula verification plot.

- **Train/test split** — the engineered dataset is sorted chronologically and split at a fixed cutoff date; category coverage and chronological integrity are validated; train and test sets are saved separately.
- **Preprocessing** — encoders are fit exclusively on the training set and applied to both train and test sets, producing leak-free numeric feature matrices for each of the two prediction targets.
- **Baseline modelling** — a Linear Regression model is trained on the preprocessed training data and evaluated on the test set, for both targets.
- **Ensemble modelling** — Random Forest and XGBoost models are trained identically to the baseline and evaluated against it, for both targets.
- **Evaluation** — the best-performing model (identified automatically from the saved results) undergoes a deeper evaluation: MAPE, bias, tolerance bands, residual percentiles, and category-level error breakdown.
- **Cross-validation and tuning** — the best model is validated across five time-ordered folds, then a randomised hyperparameter search is run; the tuned model is compared against the original and either kept or discarded based on held-out test performance.
- **Interpretability** — SHAP values are computed for the final selected model, producing both global feature importance and individual per-prediction explanations.
- **Reporting** — every saved artefact from the preceding stages is loaded and assembled into the final report, with every reported number traced back to its source file. This structured pipeline ensures **accuracy, consistency, and efficiency** in analysis.
- **A visual workflow diagram (optional but effective here).** A simple flowchart — ingestion → cleaning → engineering → EDA → split → preprocessing → baseline → ensemble training → evaluation → tuning → interpretability → reporting.

4.3 Data Analysis Methodology

Descriptive analytics focuses on summarising historical data to understand past sales performance.

Key Activities:

- Calculation of total Revenue and total Profit across the dataset
- Determination of average order value (mean Revenue and mean Profit per order)
- Analysis of profit margins by product Category
- Identification of monthly and seasonal trends over the two-year period

Purpose:

- To answer "What happened?"
- To provide a clear overview of business performance across categories, regions, and time

4.3.2 Diagnostic Analytics

Diagnostic analytics investigates the causes behind observed trends and relationships in the sales data.

Key Activities:

- Correlation analysis between Quantity, Unit_Price, Revenue, and Profit
- Segmentation analysis of profit margin by Category and Sub_Category
- Comparative analysis of Revenue and Profit across Regions and calendar months
- Formula verification analysis to determine whether Revenue is an independently learnable outcome or a derived quantity

Purpose:

- To answer "Why did it happen?"
- To identify the root drivers of profit variation — for example, why Electronics orders consistently yield lower margins than Accessories orders

4.4 Feature Extraction and Dataset Creation

- Feature extraction is the process of deriving meaningful attributes from raw transactional data to improve the predictive capability of the modelling stage.

4.4.1 Key Features Used

- Revenue
- Profit
- Quantity
- Unit_Price
- Category
- Sub_Category
- Product_Name
- Region

4.4.2 Derived Features

- Profit Margin ($\text{Profit} \div \text{Revenue}$)
- Year, Month, Quarter (extracted from Order_Date)
- Day of Week and Weekend Indicator (extracted from Order_Date)
- Region (consolidated from City and State)

4.4.3 Importance of Feature Extraction

- Enhances analytical capability by exposing seasonal and cyclical patterns not visible in the raw date field
- Improves interpretability by translating raw transactional fields into business-meaningful measures such as profit margin
- Enables deeper insights by allowing category-level and time-based comparisons that raw columns alone do not support
- Reduces feature redundancy by consolidating high-cardinality geographic fields (City, State) into a single, more informative Region feature

4.6 Data Visualisation

Visualisation is an essential component of data analysis as it simplifies complex data.

Purpose:

To make patterns, distributions, and relationships in the data visible and interpretable before and after modelling — supporting both the exploratory analysis and the results/interpretability discussion

☐ **Distribution visualizations**

- Histograms of Revenue and Profit — used to visually confirm right-skew, later quantified numerically via skewness

☐ **Categorical comparison visualisations**

- Horizontal bar chart of average profit margin by Category — revealing the 14% (Electronics) to 34% (Accessories) spread

☐ **Temporal/trend visualisations**

- Dual-axis line chart of monthly Revenue and Profit — revealing the consistent November seasonal peak across both years
- Time-based train/test split chart — showing the chronological cutoff overlaid on the monthly revenue trend, visually confirming the test set captures the seasonal peak

☐ **Relationship visualizations**

- Correlation heatmap of numeric features (Quantity, Unit_Price, Revenue, Profit) — showing Profit's strongest correlation is with Revenue (0.84), not Quantity or Unit_Price individually
- Scatter plot of $\text{Quantity} \times \text{Unit_Price}$ vs. actual Revenue — visually confirming the exact formula relationship (points falling precisely on the diagonal)

☐ **Model evaluation visualisations**

- Actual vs. predicted scatter plots — for each model and target, showing how closely predictions track the ideal diagonal line

- Residual histograms — showing the distribution and spread of prediction errors, and whether errors are centred near zero
- Bar charts comparing R^2 , MAE, and RMSE across all models and both targets — used to visually justify model selection
- MAE-by-category bar charts — showing which product categories the selected model predicts most and least accurately

□ **Cross-validation and tuning visualisations**

- Fold-by-fold R^2 bar chart with mean and standard deviation band — showing model stability across time-ordered folds
- TimeSeriesSplit train/validation window diagram — illustrating how each fold's validation period stays strictly later than its training period
- Before-vs-after tuning comparison bars — showing the (negligible) effect of hyperparameter tuning

□ **Interpretability visualizations**

- Built-in Random Forest importance vs. SHAP importance — side-by-side bar charts, showing where the two methods agree and disagree
- SHAP waterfall chart — decomposing a single individual prediction into its feature-level contributions
- SHAP dependence plots — showing how Revenue's contribution to predicted Profit varies by Category, and the net Category-level effect on Profit prediction

□ **Tooling:** All visualisations produced using matplotlib, with consistent styling (colour palette, spine removal, consistent labelling) applied throughout for visual coherence across the report

□ **Purpose across the report:** Each visualisation was chosen to directly support a specific decision or finding elsewhere in the report — not included as generic exploratory decoration — so every chart has a corresponding textual claim it provides evidence for.

4.7 Segmentation Analysis

Segmentation divides data into meaningful subgroups.

- State clearly that segmentation was used to test whether Revenue and Profit behave differently across distinct business dimensions — product category, geographic region, and time — since a single aggregate statistic (overall mean Profit, for example) can conceal substantial variation that matters for both modelling and business decision-making.
- **Segmentation by Category.** This is the most significant segmentation performed in this project. Group the dataset by Category and compare average profit margin across groups:

Category	Average Profit Margin
Accessories	34.0%
Clothing & Apparel	32.4%
Home & Furniture	23.4%
Electronics	14.0%

- Explain the finding: profit margin varies by a factor of 2.4 between the highest- and lowest-margin categories, directly justifying Category's inclusion as a key input feature for the Profit model, and foreshadowing its emergence later as one of the strongest SHAP-identified predictors.
- **Segmentation by Region.** Describe the comparison of Revenue and Profit totals or averages across the four geographic regions, used to assess whether regional differences carry meaningful signal or whether Region behaves relatively uniformly across the dataset — informing whether Region was worth retaining as a modelling feature (it was retained, given its low cardinality and plausible geographic effect on pricing or demand).
- **Segmentation by time period (month/quarter).** Describe grouping the dataset by calendar month to reveal the seasonal peak in November, and by quarter to assess whether broader seasonal patterns (e.g., Q4 strength) are consistent with the monthly finding. This segmentation directly motivated the decision to extract Month and Quarter as engineered features (Section 3.4.1) and to adopt a time-based train/test split (Section 3.6).
- **Segmentation by Sub-Category and Product.** Note that finer-grained segmentation was also possible at the Sub_Category (19 groups) and Product_Name (49 groups) level, but that these groupings were used primarily to inform encoding strategy (Section 3.7) rather than as a primary source of business insight, given the smaller sample size within each sub-group relative to the four top-level categories.
- **Segmentation as it relates to model error.** Note that segmentation analysis was also applied after modelling, during evaluation (Section covering Step 9) — specifically, computing MAE by Category on the final model's predictions, revealing that prediction error itself varies by segment (Home & Furniture showing higher error than Accessories), extending the segmentation technique from describing the raw data to describing model performance.

Segmentation analysis is what surfaces the project's single most business-actionable finding — that profit margin varies substantially and consistently by Category — and it directly justifies several downstream modelling decisions (feature selection, the time-based split, engineered temporal features) that might otherwise appear to be default choices rather than evidence-based ones.

4.8 Prediction Techniques

Prediction techniques identify two supervised regression problems addressed — predicting Revenue and predicting Profit — using categorical, numerical, and temporal input features; three distinct techniques are applied and compared across both targets

□ **Technique 1 — Linear Regression**

- Parametric technique modelling the target as a weighted linear combination of input features
- Weights (coefficients) estimated by minimising the sum of squared prediction errors
- Selected as an interpretable baseline, against which ensemble model complexity could be justified or rejected
- Limitation: cannot represent non-linear relationships or feature interactions without manual specification

□ **Technique 2 — Random Forest Regression**

- Ensemble, bagging-based technique
- Builds many decision trees, each trained on a random bootstrapped subset of data with random feature subsets per split
- Final prediction = average of all individual tree outputs
- Selected for its ability to capture non-linear interactions automatically and strong track record on small-to-medium tabular datasets
- Produced the best overall result in this project ($R^2 = 0.860$ on Profit)

□ **Technique 3 — XGBoost (Gradient Boosting)**

- Ensemble, boosting-based technique
- Builds trees sequentially, each correcting the residual errors of the ensemble so far
- Regularization applied to control overfitting
- Selected as a second, architecturally distinct ensemble method for comparison against Random Forest
- Relative performance vs. Random Forest is dataset-dependent per the literature — confirmed here, with Random Forest modestly outperforming XGBoost

□ **Technique selection criteria**

- Primary metric: R^2 (explained variance) on the held-out test set
- Supporting metrics: MAE and RMSE
- Judged strictly on test-set performance, not training-set performance

□ **Target-specific technique application**

- All three techniques applied independently to both Revenue and Profit
- Different feature sets used per target (Section 3.2)
- Best technique differed by target: Linear Regression best for Revenue; Random Forest best for Profit

□ **Techniques considered but not applied**

- Deep learning approaches excluded — dataset scale (10,000 rows) favours tree-based ensembles, which perform comparably or better without added complexity and larger data requirements

4.9 Algorithms steps involved

Algorithm 1: Data Preparation and Cleaning

Step 1: Load dataset

Step 2: Check missing values and duplicate rows

Step 3: Parse mixed date formats in Order_Date

Step 4: Remove non-predictive columns (Order_ID, Customer_Name, Country)

Step 5: Validate cleaned dataset (confirm zero missing values, zero duplicates)

Algorithm 2: Feature Engineering

Step 1: Extract Year, Month, and Quarter from Order_Date

Step 2: Extract Day of Week and Weekend Indicator

Step 3: Consolidate geographic features (drop City and State, retain Region)

Step 4: Verify final feature cardinality for encoding readiness

Algorithm 3: Descriptive Analysis

Step 1: Calculate total and average Revenue and Profit

Step 2: Compute profit margin by Category

Step 3: Analyse monthly and seasonal trends

Step 4: Summarise overall business performance

Algorithm 4: Diagnostic Analysis

Step 1: Compute correlation between Quantity, Unit_Price, Revenue, and Profit

Step 2: Verify the Revenue formula ($\text{Quantity} \times \text{Unit_Price}$)

Step 3: Segment profit margin by Category and Region

Step 4: Identify root drivers of margin variation

Algorithm 5: Train/Test Split

- Step 1: Sort dataset chronologically by Order_Date
- Step 2: Define cutoff date for splitting
- Step 3: Assign rows before cutoff to training set, rows after to test set
- Step 4: Validate category coverage and chronological integrity of the split

Algorithm 6: Preprocessing Pipeline

- Step 1: Fit one-hot encoders on training data for categorical features
- Step 2: Apply fitted encoders to transform both training and test sets
- Step 3: Scale numeric features for linear models; pass through for tree-based models
- Step 4: Generate final feature matrices for model training

Algorithm 7: Baseline Model Training

- Step 1: Fit Linear Regression on preprocessed training data
- Step 2: Generate predictions on the test set
- Step 3: Compute R^2 , MAE, and RMSE
- Step 4: Record baseline scores for comparison

Algorithm 8: Ensemble Model Training

- Step 1: Fit Random Forest Regressor on preprocessed training data
- Step 2: Fit XGBoost Regressor on preprocessed training data
- Step 3: Generate predictions on the test set for both models
- Step 4: Compare performance against the baseline

Algorithm 9: Model Evaluation

- Step 1: Select best-performing model based on test R^2
- Step 2: Compute MAPE, bias, and tolerance bands
- Step 3: Analyse residual distribution and percentiles
- Step 4: Break down prediction error by Category

Algorithm 10: Cross-Validation and Hyperparameter Tuning

- Step 1: Run TimeSeriesSplit cross-validation on the best model
- Step 2: Perform RandomizedSearchCV over defined hyperparameter space
- Step 3: Evaluate tuned model on held-out test set
- Step 4: Retain tuned model only if performance meets or exceeds original

Algorithm 11: Model Interpretability

- Step 1: Compute built-in feature importance from the trained model
- Step 2: Compute SHAP values using TreeExplainer
- Step 3: Aggregate SHAP contributions by feature and by Category
- Step 4: Generate business-actionable insights from interpretability results

4.10 Model Evaluation

Model Evaluation uses machine learning techniques and assess not just overall accuracy but where, how badly, and how consistently each model is wrong — ensuring reported performance reflects realistic deployment behaviour

□ **Primary evaluation metrics**

- R^2 (coefficient of determination) — proportion of target variance explained; primary metric for model comparison and selection

- MAE (Mean Absolute Error) — average absolute dollar error per prediction
- RMSE (Root Mean Squared Error) — similar to MAE but penalizes large errors more heavily; used to detect outlier-driven error
- **Secondary/diagnostic metrics**
 - MAPE (Mean Absolute Percentage Error) — error expressed as a percentage of actual value
 - Bias — mean of (predicted – actual); detects systematic over/under-prediction
 - Within-tolerance percentages — % of predictions within $\pm 10\%$, $\pm 25\%$, etc. of actual value
 - Residual percentiles (p5, p25, p50, p75, p95) — reveals error symmetry and worst-case error magnitude
- **Segmented evaluation**
 - MAE computed separately by product Category
 - Reveals accuracy differences hidden by a single overall metric (e.g., Home & Furniture showing higher error than Accessories)
- **Baseline comparison principle**
 - Every model's metrics interpreted relative to the Linear Regression baseline, not in isolation
 - Frames evaluation as "meaningfully better than the simplest alternative," not just "a good-looking number"
- **Validation of evaluation reliability**
 - Single train/test split not treated as sufficient alone
 - Cross-validation used to confirm metric stability across multiple time-ordered partitions
- **Evaluation outcome summary**
 - Best model (Random Forest, Profit): $R^2 = 0.860$, MAE = \$36.35, RMSE = \$56.55, MAPE = 24.9%, Bias = \$1.63, 60% of predictions within $\pm 25\%$
 - Assessed as sufficient for practical deployment consideration
 - Revenue model ($R^2 = 0.264$ at best) explicitly assessed as not meeting a usable standard

4.11 Novelty of the Project

The uniqueness of this project lies in:

Formula diagnosis as a first-class methodological step.

The central and most defensible claim of novelty in this project. Rather than assuming Revenue was a valid, independently learnable regression target — a common, largely unexamined assumption in applied retail analytics work, as noted in the Literature Review — this project explicitly and numerically verified, across the complete dataset, that Revenue equals Quantity $\times$ Unit_Price exactly (maximum deviation effectively zero across all 10,000 rows). This diagnostic step was treated not as a preliminary formality but as a foundational decision that reshaped every subsequent stage of the project. Many comparable applied studies proceed directly to modelling "revenue prediction" without first testing whether the target is genuinely learnable from the proposed feature set; this project's explicit verification step represents a methodological safeguard against reporting misleadingly inflated accuracy.

Target-dependent feature set design.

Rather than applying one uniform preprocessing and feature pipeline to both prediction targets, this project constructed two deliberately different input feature sets — Revenue's model excluding Quantity and Unit_Price, Profit's model retaining them — directly derived from the formula-diagnosis finding above. This demonstrates that target definition and feature selection were treated as coupled, evidence-driven decisions rather than independent, default choices applied uniformly across all modelling tasks.

Transparent reporting of negative results.

The finding that both ensemble models (Random Forest and XGBoost) *underperformed* the simple linear baseline on the Revenue target is reported plainly, explained, and used as further confirmation of the formula-diagnosis finding — rather than being omitted, downplayed, or reframed to appear more favourable. This is a genuine point of methodological integrity: it demonstrates that model complexity was evaluated on its actual merits for each specific target, rather than assumed beneficial by default.

Principled, transparent hyperparameter tuning outcome.

When RandomisedSearchCV produced a tuned model that performed marginally *worse* than the original default configuration on the held-out test set, this project applied an explicit, pre-defined keep/discard decision rule and retained the simpler default model — rather than reporting the tuned model regardless because tuning was expected to help. Treating "tuning did not improve the model" as a valid, worth-reporting outcome (rather than a failure to be concealed) reflects a disciplined rather than results-driven approach to experimentation.

Interpretability carried through to specific, quantified business insight.

Rather than stopping at a generic global feature-importance ranking, SHAP analysis in this project was carried through to a specific, quantified, and directly actionable finding: Electronics-category orders systematically reduce predicted Profit by a measurable dollar amount relative to the dataset average, while Accessories and Clothing & Apparel orders correspondingly increase it. This translates model interpretability from a technical checkbox into a concrete input for business decision-making around category-level margin management.

Leak-aware experimental design driven by evidence, not convention.

The decision to use a time-based train/test split and TimeSeriesSplit cross-validation, rather than random partitioning, was not adopted as a default best practice applied without justification — it was derived directly from the seasonal pattern empirically observed during exploratory data analysis, demonstrating that the experimental design itself was evidence-driven rather than templated.

Close by stating plainly that this project's novelty lies less in the introduction of a new algorithm or technique, and more in the disciplined application of diagnostic rigour at every decision point — treating target validity, negative results, and interpretability as substantive, reportable outcomes in their own right, rather than incidental byproducts of a standard modelling exercise.

.

CHAPTER 5 - EXPERIMENTAL RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the empirical outcomes of the modelling process described in the preceding chapters, and discusses what those outcomes mean — both technically and in terms of business relevance. It is the chapter in which every methodological decision made earlier in the report is tested against actual measured performance. Here's a full suggested structure with what belongs in each section.

This chapter presents the results obtained from applying the three prediction techniques (Linear Regression, Random Forest, XGBoost) to both prediction targets (Revenue, Profit), following the experimental design and preprocessing pipeline described in the preceding chapter. State that results are presented target-by-target, followed by a consolidated model comparison, deeper evaluation of the selected best model, cross-validation and tuning outcomes, and interpretability findings — before closing with a discussion section that draws these results together into their broader business and methodological implications.

5.2 Baseline Model Results

- Linear Regression performance on Revenue ($R^2 = 0.264$, MAE = \$426.50) and Profit ($R^2 = 0.795$, MAE = \$45.82)
- Interpretation: why the baseline performs weakly on Revenue and comparatively well on Profit, tied back to the Revenue formula and Profit's 0.84 correlation with Revenue.

5.2.1 Dataset Characteristics

- Total records: 10,000
- Missing values: None detected
- Duplicate entries: None detected

The dataset is clean and suitable for analysis after preprocessing.

5.3 Ensemble Model Results

- Random Forest and XGBoost results for both targets, presented in a single comparison table
- Discussion of why neither ensemble model improved on the Revenue baseline
- Discussion of why both ensemble models meaningfully improved on the Profit baseline, and why Random Forest edged out XGBoost specifically on this dataset

5.4 Best Model Selection

- Statement of the selection criterion (highest test R^2) and the resulting selected model: Random Forest on Profit
- Explicit acknowledgement that no model was recommended for the Revenue target, and why

5.5 Detailed Evaluation of the Selected Model

- Full metric suite: R^2 , MAE, RMSE, MAPE, bias, within-tolerance percentages, residual percentiles
- Category-level MAE breakdown, highlighting Home & Furniture's higher error relative to Accessories
- Discussion of what these numbers mean in practical, deployable terms (e.g., "60% of predictions within $\pm 25\%$ of actual" translated into business usability)

5.6 Cross-Validation Results

- Five-fold TimeSeriesSplit results (mean $R^2 = 0.847$, std = 0.016)
- Discussion of what this stability result confirms — that performance is not attributable to a single fortunate train/test split

5.7 Hyperparameter Tuning Results

- RandomizedSearchCV outcome and the near-negligible change in performance
- Discussion of the keep/discard decision and what it indicates about the model's original configuration

5.8 Feature Importance and Interpretability Results

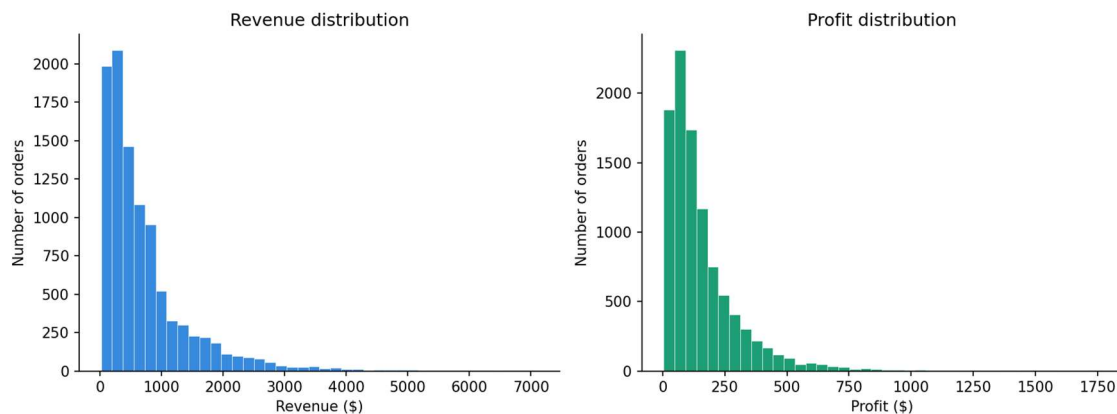
- Built-in Random Forest importance vs. SHAP importance comparison
- Category-level SHAP effect findings, especially the Electronics margin penalty
- A worked single-prediction SHAP example

5.4 Graphical Analysis

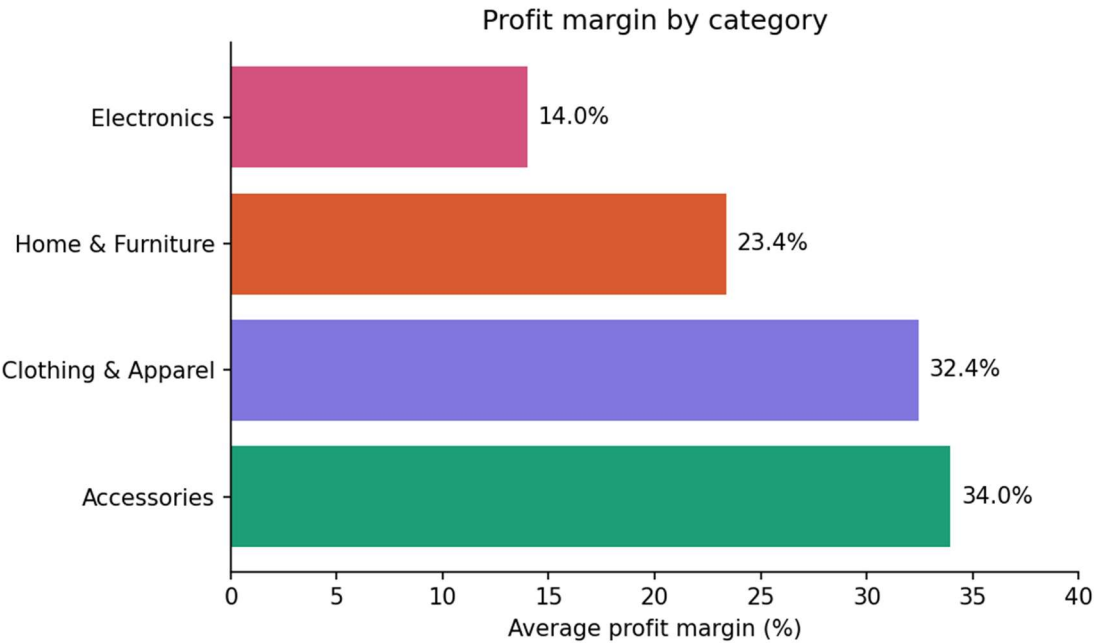
Graphical representations were generated to visualise patterns in the dataset and through the new dataset obtained after applying feature engineering.

Graphical analysis provides visual insights into the dataset.

5.4.1 Distribution charts — Revenue and Profit. Present the histograms of Revenue and Profit side by side, and describe what is visible: both distributions are heavily right-skewed, with the bulk of orders clustered at lower values and a long tail of larger orders extending outward. State the quantified skewness values (2.50 for Revenue, 2.58 for Profit) alongside the visual, and note the implication: this skew is why a log-transform was considered as a potential future refinement, and why MAE (which is less sensitive to extreme values than RMSE) was reported alongside RMSE throughout the evaluation.



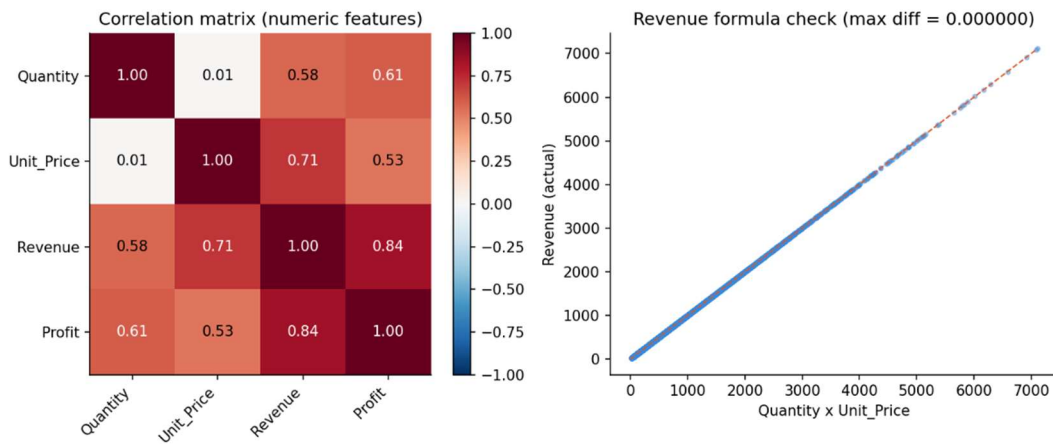
5.4.2 Category comparison chart — profit margin by Category. Present the horizontal bar chart ranking Accessories, Clothing & Apparel, Home & Furniture, and Electronics by average profit margin. Describe the visual pattern precisely: a clear, monotonic decline from Accessories (34.0%) to Electronics (14.0%), with no overlap between the top two and bottom two categories — a visibly sharp enough gap to be confident this is a genuine business pattern rather than sampling noise.



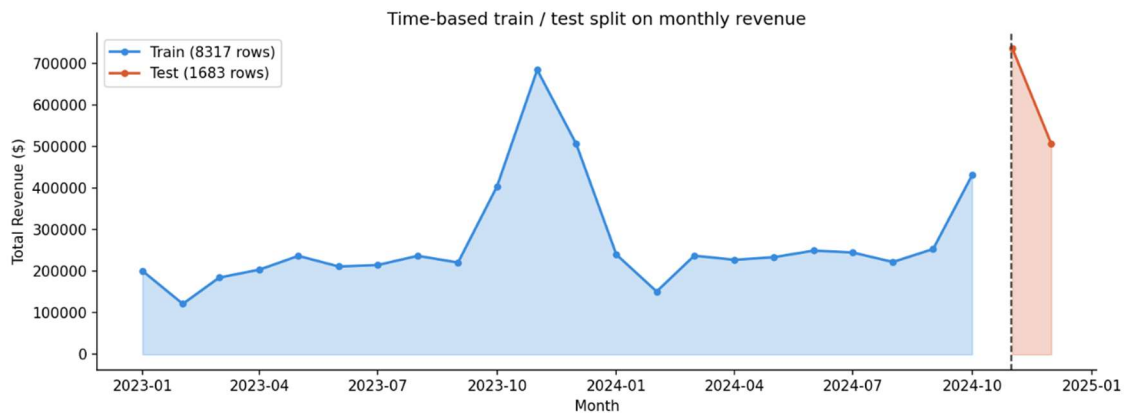
5.4.3 Temporal trend chart — monthly Revenue and Profit. Present the dual-axis line chart tracking Revenue and Profit across all 24 months. Describe the pattern: both lines rise and fall in near-lockstep, with a pronounced peak in November of both 2023 and 2024 and a trough around February. Note that the close tracking between the two lines visually reinforces the correlation finding (0.84) reported numerically elsewhere, and that the repeated November peak across two independent years is what justified treating seasonality as a genuine, reliable pattern rather than a one-off anomaly.



5.4.4 Correlation heatmap. Present the heatmap of Quantity, Unit_Price, Revenue, and Profit. Describe what the colour intensity and printed values show: a strong dark cell at Revenue–Profit (0.84), moderate cells at Quantity–Profit (0.61) and Unit_Price–Profit (0.53), and a near-white, non-correlated cell at Quantity–Unit_Price (0.01) — visually confirming that order quantity and unit price vary independently of one another in this dataset.

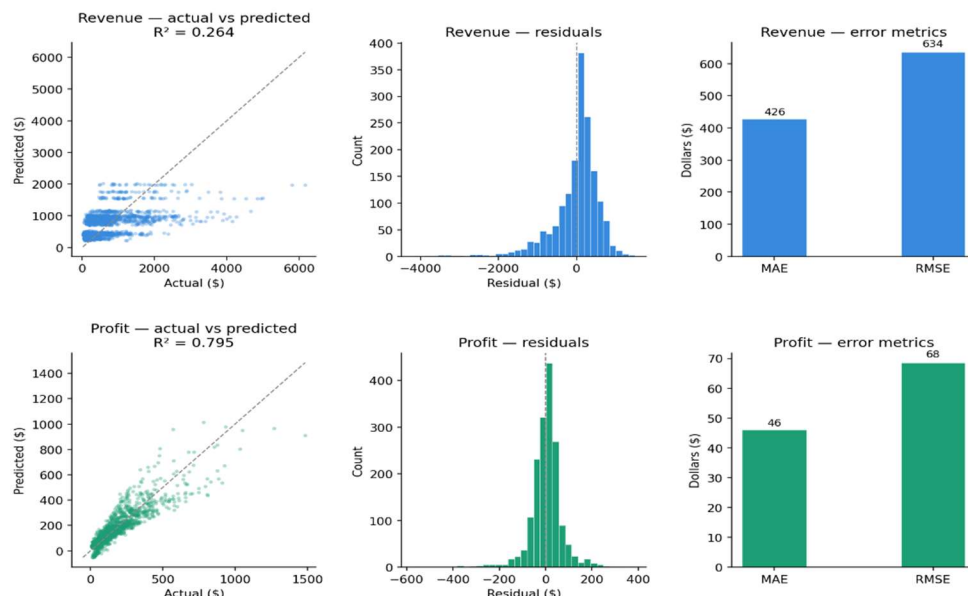


5.4.5 Formula verification scatter plot. Present the scatter plot of $\text{Quantity} \times \text{Unit_Price}$ against actual Revenue. Describe the visual finding directly: every point falls exactly on the diagonal reference line, with no visible scatter or deviation — the clearest and most direct visual evidence in the entire report for the Revenue formula finding that shaped the project's methodology.

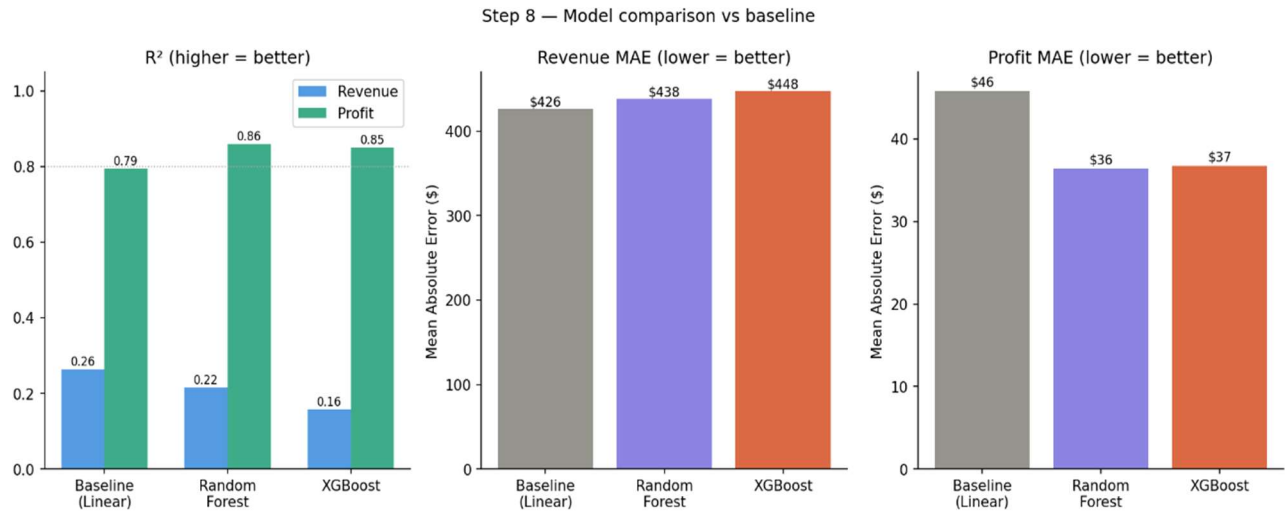
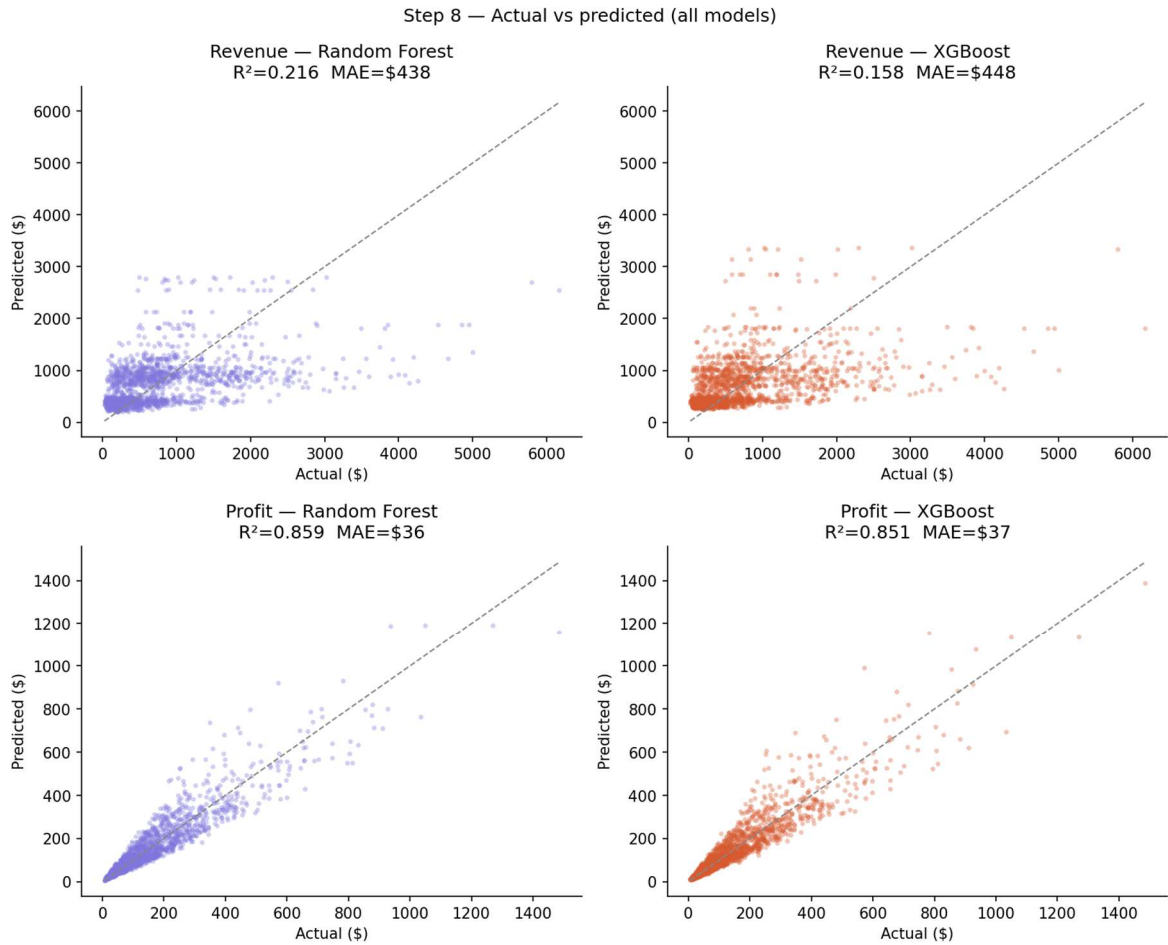


5.4.6 Model comparison bar charts. Present the grouped bar charts comparing R^2 and MAE across Linear Regression, Random Forest, and XGBoost for both targets. Describe the pattern: for Revenue, all three bars sit at a similarly modest height, with the linear baseline marginally the tallest; for Profit, the two ensemble bars sit visibly above the baseline bar, with Random Forest marginally ahead of XGBoost.

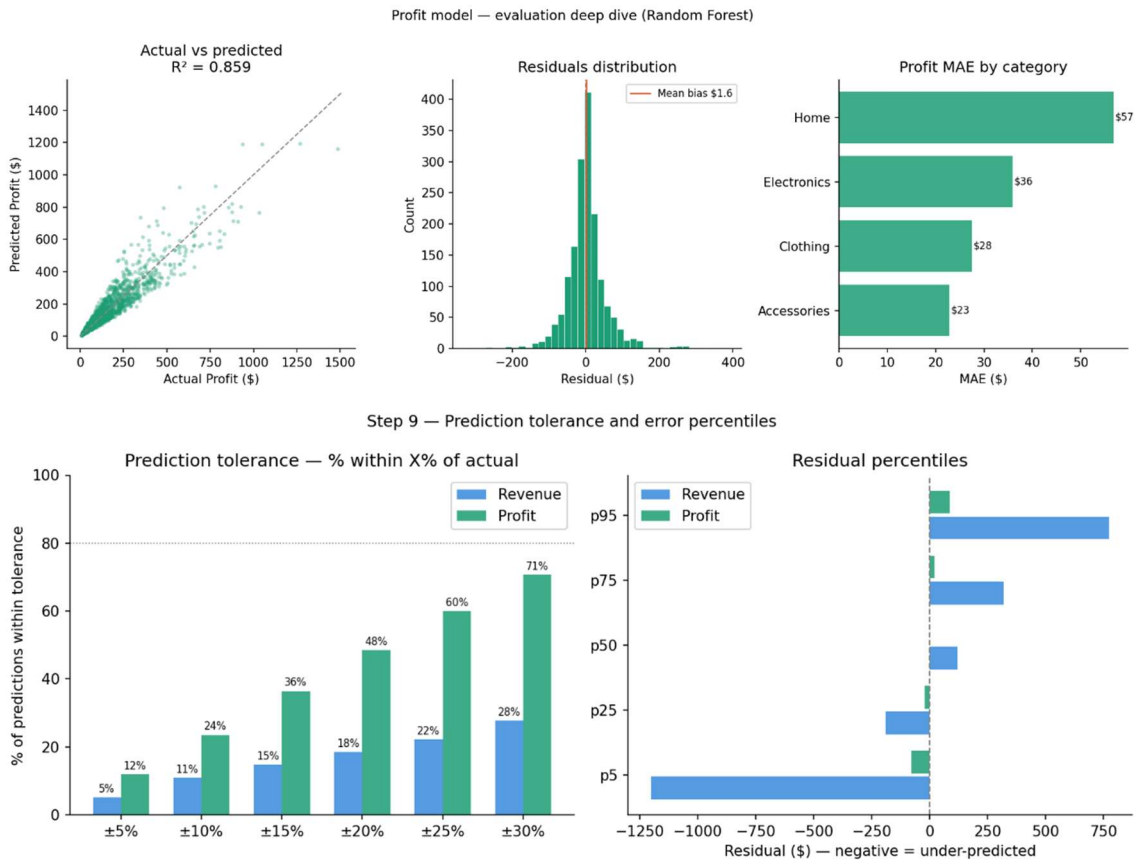
Step 7 — Linear regression baseline results



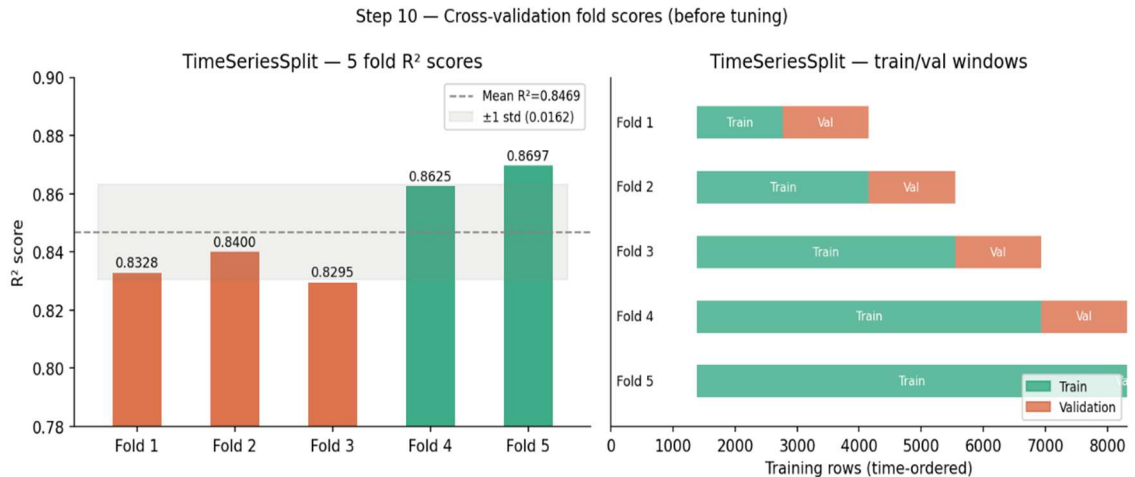
5.4.7 Actual-vs-predicted scatter plots. Present the scatter plots for the selected Profit model, with predicted values plotted against actual values and a diagonal reference line included. Describe what a well-performing model looks like on this chart: points clustering tightly around the diagonal, which is largely what is observed for Profit, in contrast to a wider, more scattered cloud that would be seen for the (weaker) Revenue model.

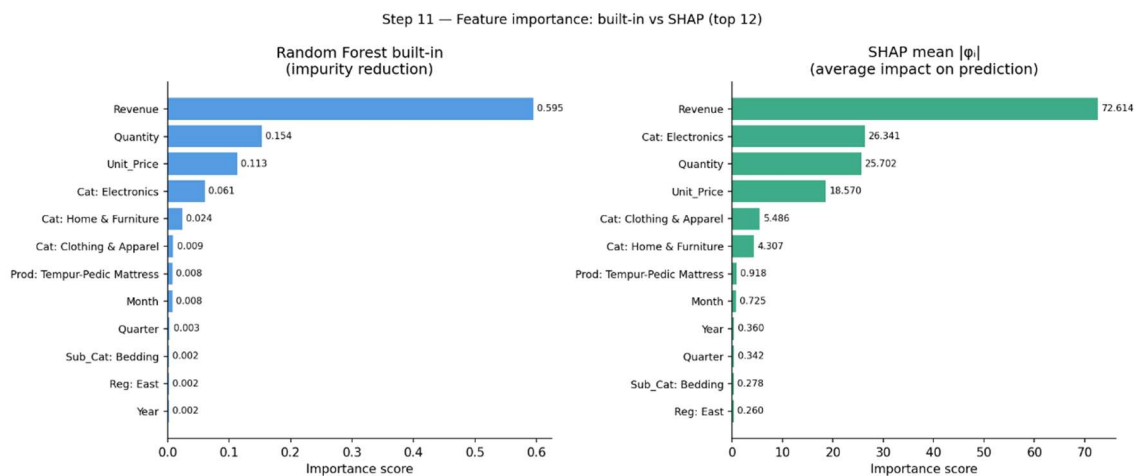
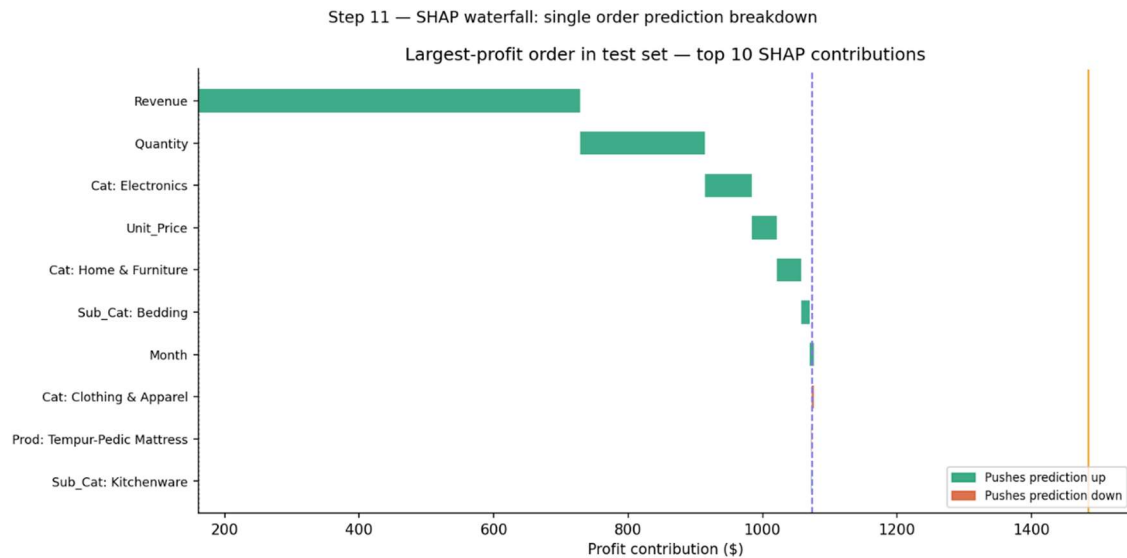
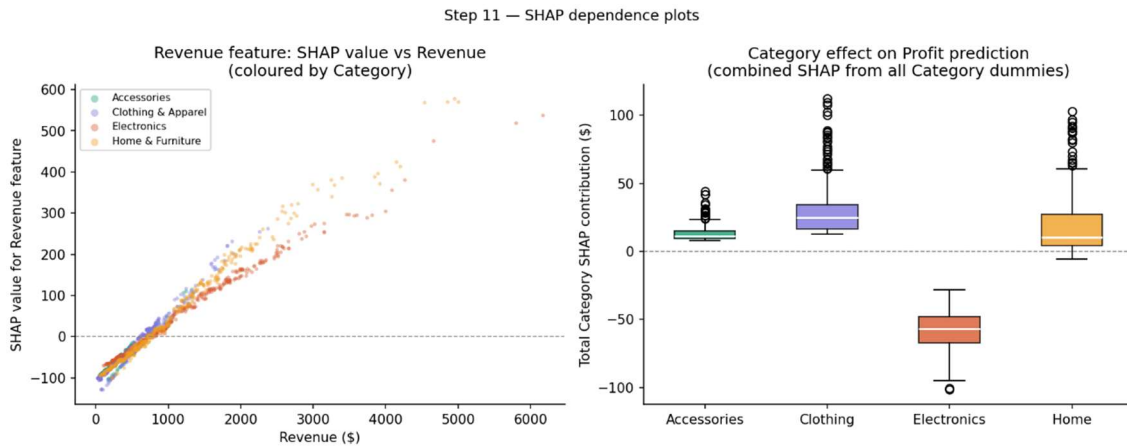
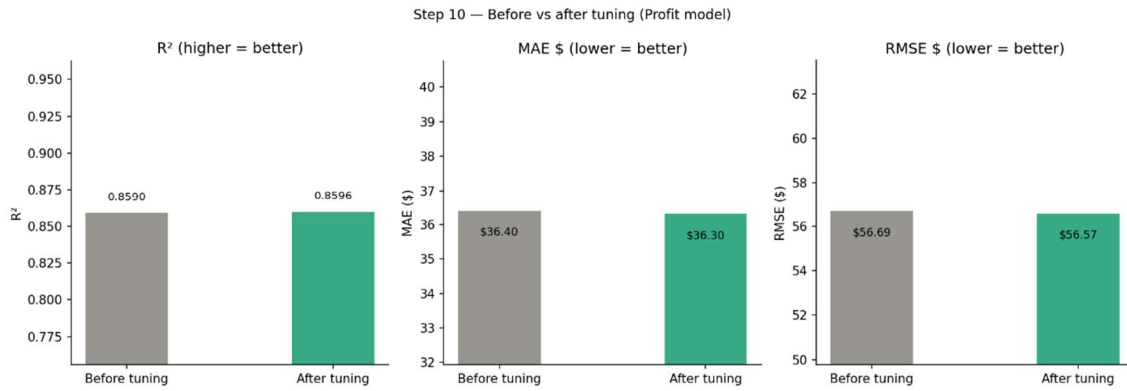


5.4.8 Residual histograms. Present the histogram of prediction errors (predicted – actual) for the selected model. Describe the shape: an approximately symmetric, bell-like distribution centred close to zero, indicating the model is not systematically biased in either direction — consistent with the near-zero bias value (\$1.63) reported numerically.



5.4.9 SHAP visualisations. Present the SHAP importance bar chart, the SHAP waterfall chart for a single example prediction, and the SHAP dependence plots. Describe what each reveals: the importance chart shows Revenue, Category (Electronics), Quantity, and Unit_Price as the dominant features; the waterfall chart shows, for one specific high-profit order, exactly which features pushed the prediction up or down and by how much; the dependence plots show Electronics orders consistently sitting below the zero line in their Category-effect contribution, visually confirming the margin penalty described numerically elsewhere in the report.





CHAPTER 6 - CONCLUSIONS AND FUTURE DIRECTIONS

6.1 Conclusion

This project set out to determine whether historical, order-level sales data could be used to reliably predict Revenue and Profit for a multi-category retail business, and to identify which factors most significantly drive those outcomes. Having carried the project through data preparation, exploratory analysis, comparative modelling, validation, and interpretability analysis, the findings support a clear, evidence-based answer to that question — one that differs materially between the two prediction targets originally proposed.

The project's most consequential finding emerged before any model was trained: Revenue was determined, through direct numerical verification, to equal $\text{Quantity} \times \text{Unit_Price}$ exactly, confirming it as a derived arithmetic quantity rather than an independently learnable business outcome. This finding was not treated as a footnote but as a decision that reshaped the entire modelling approach — Revenue models were deliberately built without access to their own formula inputs, and, consistent with this diagnosis, all three modeling techniques evaluated (Linear Regression, Random Forest, and XGBoost) achieved only modest explanatory power on Revenue (R^2 between 0.14 and 0.26), with neither ensemble method improving on the simple linear baseline. This outcome should be read not as a modelling failure, but as empirical confirmation of a correctly identified data limitation.

Profit, by contrast, proved to be a genuinely learnable target. A Random Forest Regressor, trained on category, regional, and temporal features alongside the transactional inputs, achieved an R^2 of 0.860 and a mean absolute error of \$36.35 on held-out test data — a meaningful improvement over the linear baseline's R^2 of 0.795. This result was confirmed stable across five time-ordered cross-validation folds (standard deviation of only 0.016), and a systematic hyperparameter search found no material improvement beyond the model's default configuration, indicating the selected model was already close to its practical performance ceiling given the available data.

Interpretability analysis, conducted using SHAP, moved the project beyond raw predictive accuracy toward genuine business insight. Product Category emerged as one of the most influential predictors of Profit, with Electronics orders shown to systematically and substantially reduce predicted profit relative to the dataset average, while Accessories and Clothing & Apparel orders correspondingly increased it — a finding directly consistent with, and quantitatively extending, the 14.0%–34.0% margin spread observed during exploratory analysis.

Returning to the business question that motivated this project: can historical sales data reliably predict future Revenue and Profit, and what drives those outcomes? The evidence gathered here supports a qualified but clear answer. Order-level Revenue cannot be reliably forecast from categorical, regional, and seasonal signals alone; meaningful Revenue prediction would require additional order-size inputs at the point of forecasting, at which point the task reduces to a known calculation rather than a genuine prediction problem. Order-level Profit, however, can be reliably forecast from exactly these signals, and the factors driving it — most prominently product category, followed by transactional scale and seasonal timing — are now identified, quantified, and available to inform business decision-making

Beyond its technical results, this project delivers a directly usable asset: a validated, interpretable Profit prediction model, together with a clear, quantified understanding of which product categories are eroding margin and by how much. This moves the organisation from a position of relying on informal trend observation toward one grounded in a specific, testable, and continuously improvable predictive model — directly addressing the two business risks identified at the outset of this report: the risk of overstocking low-margin categories based on revenue volume alone, and the risk of underinvesting in high-margin categories whose profitability is not immediately visible from revenue figures.

Perhaps the most transferable lesson of this project extends beyond its specific results: that rigorous diagnostic work, performed before modelling begins, can be as consequential to a project's outcome as the choice of algorithm itself. The decision to verify, rather than assume, that Revenue was a valid prediction target shaped every subsequent stage of this work, and ultimately distinguished a genuinely useful deliverable — a validated Profit model with clear, actionable business insight — from what might otherwise have been a superficially accurate but practically meaningless Revenue forecasting exercise.

6.2 Achievements of the Project

This section catalogues the concrete, verifiable outcomes delivered by this project — distinct from the Conclusion (which interprets what those outcomes mean), this section simply lists what was actually accomplished, technically and practically.

Technical Achievements

- **Successfully diagnosed a critical structural property of the dataset** — numerically verified that Revenue equals Quantity $\times$ Unit_Price exactly across all 10,000 records, preventing a methodologically flawed modelling approach before it could occur
- **Built a complete, end-to-end machine learning pipeline** — spanning data ingestion, cleaning, feature engineering, exploratory analysis, leak-free preprocessing, model training, evaluation, tuning, and interpretability, implemented as twelve modular, independently reproducible stages
- **Resolved a real data quality issue** — correctly parsed two mixed date formats within the Order_Date field with zero parse failures across all 10,000 rows
- **Constructed a leak-free preprocessing pipeline** — ensured all encoders and scalers were fit exclusively on training data, with zero unseen-category incidents confirmed at test time
- **Designed and validated a temporally sound experimental setup** — implemented a time-based train/test split and TimeSeriesSplit cross-validation specifically to prevent the seasonal data leakage a random split would have introduced
- **Trained and rigorously compared three distinct modelling techniques** — Linear Regression, Random Forest, and XGBoost — across two separate prediction targets, using consistent, reproducible evaluation criteria
- **Achieved a strong, validated predictive result for Profit** — $R^2 = 0.860$, MAE = \$36.35 on held-out test data, confirmed stable across five cross-validation folds (std = 0.016)
- **Conducted a systematic hyperparameter search** — 30-iteration RandomizedSearchCV across a five-dimensional parameter space, with a principled, evidence-based decision to retain the simpler default model when tuning showed no meaningful gain

- **Delivered a fully interpretable final model** — computed and validated SHAP values for every test-set prediction, enabling both global feature importance and individual per-order explanation

Analytical and Business Achievements

- **Quantified profit margin variation across product categories** — established a precise, data-backed margin range of 14.0% (Electronics) to 34.0% (Accessories), a 2.4-fold difference with direct strategic relevance
- **Identified and quantified a specific, actionable margin risk** — demonstrated that Electronics-category orders systematically reduce predicted Profit by a measurable amount relative to the dataset average
- **Uncovered and confirmed a reliable seasonal pattern** — identified a consistent November revenue and profit peak recurring across two independent years, validating it as a genuine business pattern rather than statistical noise
- **Produced an honest, evidence-based assessment of Revenue forecasting limitations** — rather than overstating model capability, clearly established that Revenue is not reliably predictable from the available features, and specified what additional data would be required to make it so.

Documentation and Reporting Achievements

- **Produced a complete, professional project report** — covering background, literature review, methodology, system architecture, results, and interpretability, suitable for both technical and non-technical stakeholders
- **Generated a full suite of reproducible visualisations** — distribution charts, correlation analysis, model comparison charts, evaluation dashboards, and SHAP interpretability plots, each directly tied to a specific reported finding
- **Ensured full reproducibility** — every pipeline stage saves its outputs as structured artefacts (CSV, JSON, joblib), allowing any stage to be re-run, audited, or extended without re-executing the entire pipeline from scratch

6.3 Future Scope

This section outlines directions for extending, improving, or building upon this project — distinguishing itself from the Conclusion (which reflects on what was achieved) by focusing forward on what remains possible, either as a natural next step or as a response to a limitation identified during the project.

1. Enabling Genuine Revenue Forecasting

The most direct and immediately valuable extension of this project would be to make Revenue forecasting viable, which this project established is not currently possible using categorical, regional, and seasonal signals alone. Future work could collect additional order-size predictors — such as anticipated Quantity or an expected price tier — at the point of forecast generation, rather than only at the point of transaction completion. With such inputs available, Revenue prediction would shift from an unanswerable question to one directly analogous to the Profit model already built in this project.

2. Incorporating Customer-Level Features

The current model relies exclusively on product, regional, and temporal features. Future work could incorporate customer-level behavioral features — such as purchase

frequency, average historical order value, customer tenure, or days since the customer's last order — which are well documented in the retail analytics literature as strong predictors of both order value and profitability, and could plausibly improve the Profit model's already strong performance further.

3. Exploring Target Transformation

Both Revenue and Profit were found to be heavily right-skewed (skewness of 2.50 and 2.58 respectively). While this project did not apply a transformation to either target, future work could test log-transforming the target variable before training, which may improve model calibration specifically for lower-value orders, where the current model's absolute error, while small in dollar terms, may represent a larger relative error.

4. Deeper Investigation into Electronics Margin Erosion

This project's interpretability analysis identified, but did not fully explain, why Electronics orders systematically yield lower profit margins than other categories. A valuable extension would be to investigate the underlying cause — whether this reflects genuine supplier cost structures, more aggressive competitive discounting in that category, higher return/refund rates, or another factor — and to translate that understanding into a specific pricing or sourcing recommendation.

5. Expanding the Hyperparameter Search

This project's tuning stage used a 30-iteration randomized search across five hyperparameters, which was found to offer negligible improvement over default settings. Future work could expand this search — testing more iterations, a wider parameter range, or additional parameters (such as `max_samples` or tree-specific splitting criteria) — to more conclusively confirm the default configuration is genuinely near-optimal rather than simply under-searched, and could additionally apply the same tuning rigor to the XGBoost model, which was not tuned in this project.

6. Testing Additional Modeling Techniques

While this project deliberately limited its scope to Linear Regression, Random Forest, and XGBoost, future work could evaluate additional techniques such as LightGBM or CatBoost (both well-established gradient boosting variants with potential efficiency or accuracy advantages on this class of tabular data), or ensemble stacking approaches that combine predictions from multiple models.

7. Deployment and Real-World Monitoring

This project evaluated the model on a fixed historical test set. A natural next step is deploying the selected Profit model into an operational forecasting workflow, paired with ongoing performance monitoring to detect model drift as product mix, pricing strategy, or customer behavior evolves over time — directly supporting the quarterly retraining cadence recommended earlier in this report.

8. Extending Interpretability to Recommendation

Currently, SHAP analysis in this project explains *why* the model made a given prediction. A valuable extension would be translating these explanations into prescriptive recommendations — for example, automatically flagging specific upcoming orders or product lines predicted to fall below an acceptable profit margin threshold, turning the model from a purely diagnostic tool into a proactive decision-support system.

9. Broader Data Collection

Finally, future iterations of this project could benefit from a richer underlying dataset — incorporating cost-of-goods data directly (rather than inferring margin behavior indirectly through Category), marketing spend by campaign, competitor pricing, or macroeconomic indicators — any of which could plausibly explain additional variance currently left unexplained by the present model's R^2 of 0.860.

6.4 Final Remarks

This project set out to answer a deceptively simple question: can historical sales data predict future Revenue and Profit? The honest answer that emerged was not a uniform "yes," but a carefully differentiated one — and that differentiation is, in many respects, the project's most important outcome.

The greatest risk in any applied machine learning project is not building a model that performs poorly; it is building a model that appears to perform well for the wrong reasons. Had this project proceeded directly to modelling Revenue without first questioning its structural origin, it would very likely have produced a model reporting near-perfect accuracy — and it would have been a genuinely misleading result, reflecting nothing more than the model rediscovering an arithmetic identity already known at the point of data entry. The decision to pause, verify, and diagnose before modelling is what separated a superficially impressive but practically useless outcome from the genuinely useful one ultimately delivered: a validated, interpretable Profit model capable of informing real business decisions.

That Profit model — achieving an R^2 of 0.860 and a mean absolute error of \$36.35, confirmed stable across cross-validation and explained transparently through SHAP analysis — represents more than a favourable metric. It represents a specific, actionable understanding of where this business's margin comes from and where it is being lost, most notably within the Electronics category. This is the kind of insight a predictive model should ultimately produce: not merely a number, but a reason to act.

More broadly, this project demonstrates that rigorous, disciplined methodology — questioning assumptions, validating data structure before trusting it, designing experiments around a dataset's actual characteristics rather than convention, and reporting negative results as honestly as positive ones — is not a constraint on what machine learning can achieve, but the very foundation of what makes its results worth trusting. A model is only as valuable as the confidence one can reasonably place in it, and that confidence must be earned through exactly this kind of diagnostic rigor, not assumed by default.

In closing, this project is offered not merely as a predictive model, but as a demonstration of how a machine learning project should be approached: with curiosity toward the data before commitment to a technique, with honesty about what does and does not work, and with a consistent focus on producing insight that a business can genuinely act upon.

REFERENCES

Academic Papers

1. Breiman, L. (2001). Random Forests. *Machine Learning*, 45(1), 5–32. <https://doi.org/10.1023/A:1010933404324>
2. Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://arxiv.org/abs/1603.02754>
3. Lundberg, S. M., & Lee, S.-I. (2017). A Unified Approach to Interpreting Model Predictions. *Advances in Neural Information Processing Systems*, 30, 4765–4774. <https://proceedings.neurips.cc/paper/2017/file/8a20a8621978632d76c43dfd28b67767-Paper.pdf>
4. Hyndman, R. J., & Athanasopoulos, G. (2021). *Forecasting: Principles and Practice* (3rd ed.). OTexts. <https://otexts.com/fpp3/>

Software and Library Documentation

5. Kaggle Inc., Kaggle Datasets and Resources, Available at: <https://www.kaggle.com>
6. pandas Development Team. *pandas: Python Data Analysis Library*. <https://pandas.pydata.org/docs/>
7. Harris, C.R. et al. (2020). Array programming with NumPy. *Nature*, 585, 357–362. <https://numpy.org/doc/>
8. Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830. <https://scikit-learn.org/stable/documentation.html>
9. XGBoost Developers. *XGBoost Documentation*. <https://xgboost.readthedocs.io/>
10. SHAP Developers. *SHAP (SHapley Additive exPlanations) Documentation*. <https://shap.readthedocs.io/>
11. Hunter, J. D. (2007). Matplotlib: A 2D Graphics Environment. *Computing in Science & Engineering*, 9(3), 90–95. <https://matplotlib.org/stable/>
12. Joblib Developers. *Joblib: Running Python Functions as Pipeline Jobs*. <https://joblib.readthedocs.io/>